

B.Tech. II Sem.
Physics-PH113

“Newton’s Ring ”

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Recapitulation

- Redistribution of intensity in space due to superposition of two or more waves of nearly same amplitude, same frequency and having constant phase difference is known as interference of light.
- The two sources are said to be coherent if the waves emitted by these sources have same frequency and phase difference between them remains constant with time.
- Coherent sources can be either produced by division of wave front or division of amplitude.
- When a wave is reflected by optically dense medium, it undergoes a phase change of π and a path difference of $\lambda/2$. This is known as Stoke's law.
- In thin film interference the two coherent sources are produced by division of amplitude.
- Interference fringe patterns obtain by reflected and transmitted light in thin film interference are complimentary of each other.

Types of Fringes Formed in Thin Film

Fringes of equal inclination or Haidinger fringes:

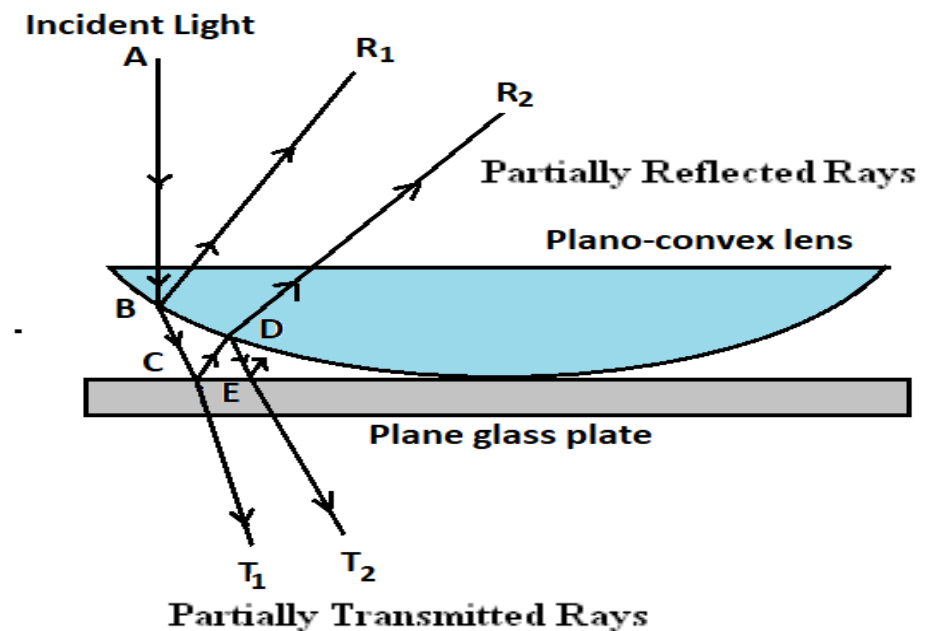
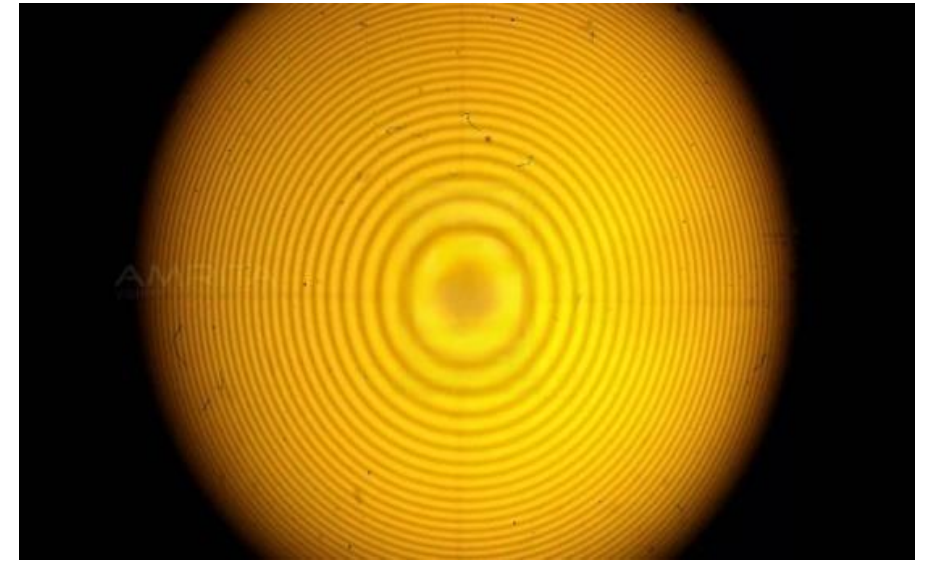
- These type of fringes are formed when a thin film of uniform thickness is illuminated by monochromatic light .
- In this case μt and λ are constant.
- So the fringes are locus of those points where angle of inclination is same.
- Hence they are also known as fringes of inclination.
- These fringes are circular in shape.
- These fringes are non localized.
- Example: Circular fringes in Michelson Interferometer.

Fringes of equal width or Fizeau fringes:

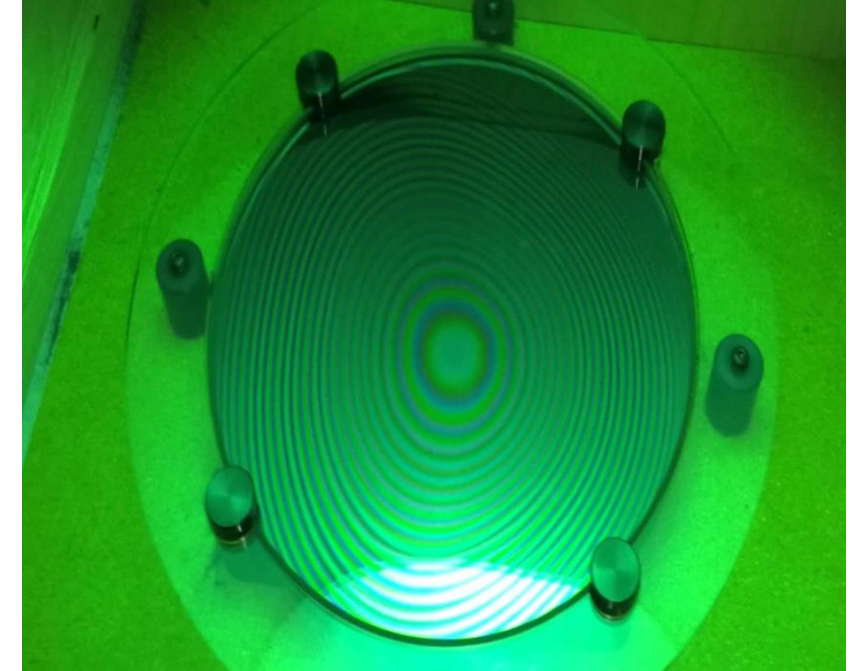
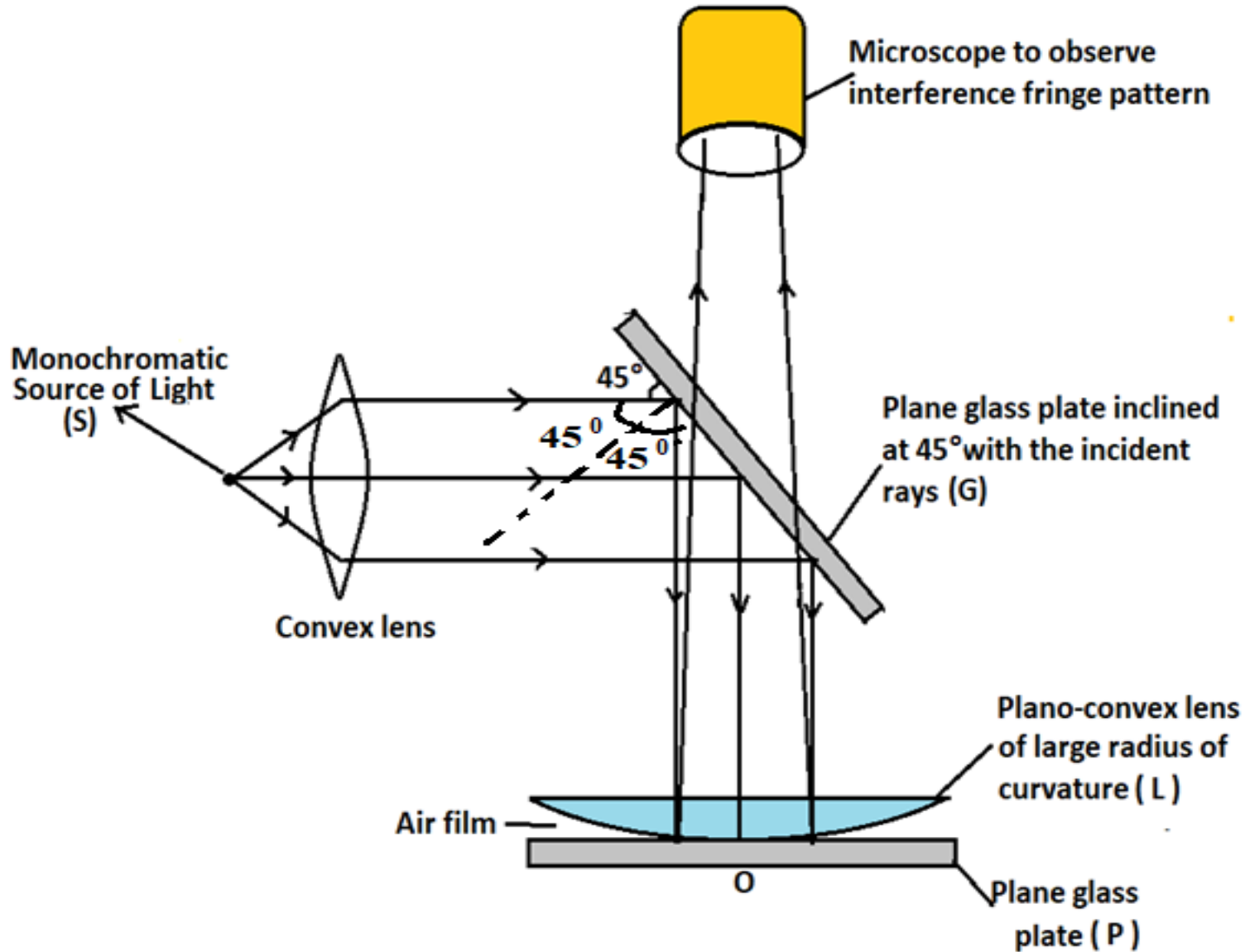
- These type of fringes are formed when a thin film of variable thickness is illuminated by monochromatic light.
- In this case angle of incidence is kept constant and λ is constant.
- So, fringes originate for different values of μt and n i.e. these are locus of those points where $\mu t = \text{constant}$.
- Hence they are also known as fringes of equal thickness.
- The shape of fringes depends on the geometry of film.
- These fringes are localized.
- Example: Fringes obtained in Newton's ring experiment.

Newton's Rings

- An air film of gradually increasing thickness is formed between the convex surface of plano-convex lens and plane glass plate.
- At point of contact thickness of film is zero and increases as one moves away.
- If monochromatic light is allowed to fall normally and the film is viewed in reflected light, alternate dark and bright rings concentric around the point of contact are seen.
- The loci of constant thickness are circles with point of contact as centre and therefore circular fringes are formed when this arrangement is illuminated at normal incidence.
- Newton's rings are formed due to the superposition of light rays obtained by the division of amplitude.



Newton's Ring Experimental Arrangement



Newton's rings

Source: <https://101viva.wordpress.com/2015/02/17/newtons-ring-experiment/>

Prepared By Dr. Dheeraj Bhardwaj

Newton's Ring Experimental Arrangement



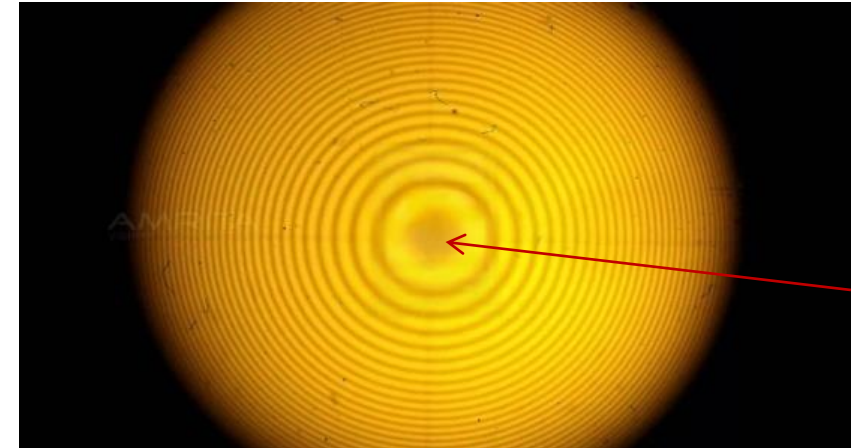
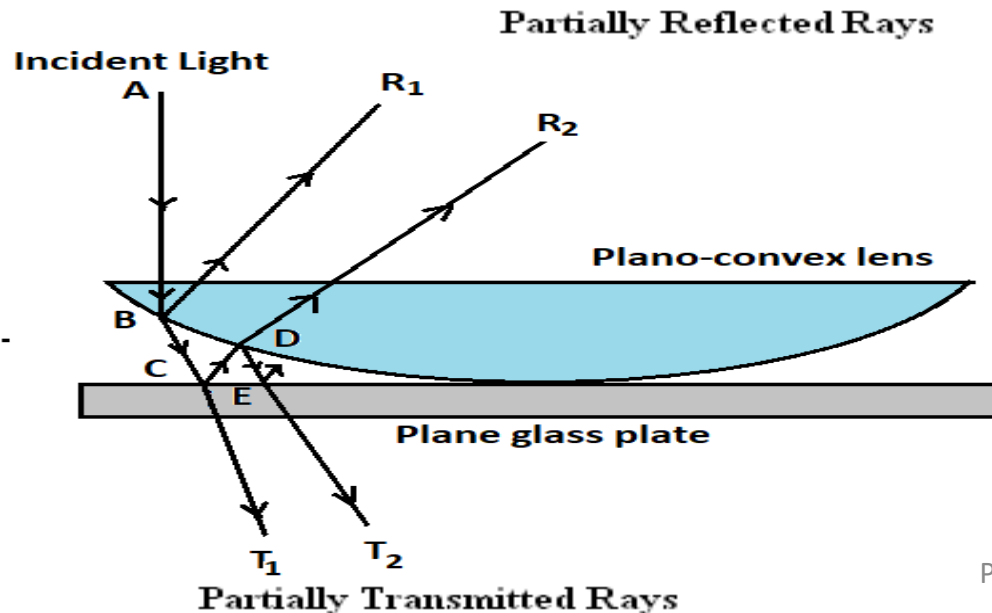
Newton's Ring in Reflected Light

The path difference between the two partially reflected rays is

$$\Delta_{eff} = 2\mu t \cos(r + \theta) + \frac{\lambda}{2}$$

As radius of curvature of plano convex lens is very large so, neglect θ . For air film $\mu = 1$ and for normal incidence $r = 0$
 $\Rightarrow \cos r = 1$ so, the effective path difference

$$\Delta_{eff} = 2t + \frac{\lambda}{2} \quad \dots(1)$$



Dark
center

Source: <http://www.amrita.edu/create>

Central Spot: At the point of contact between the lens and the plate, $t = 0$,

$$\Delta_{eff} = \frac{\lambda}{2} \rightarrow \text{condition for destructive interference}$$

So, the central fringe is always dark for reflected light.

Newton's Ring in Reflected Light..

Condition for bright rings:

General condition; path difference = $n\lambda$

$$\Delta_{eff} = 2\mu t + \frac{\lambda}{2} = n\lambda$$

[If the film enclosed has a refractive index μ]

$$2\mu t = (2n - 1)\frac{\lambda}{2} \quad \dots(2)$$

For air film $\mu=1$

$$\Rightarrow 2t = (2n - 1)\frac{\lambda}{2}$$

Where $n = 1, 2, 3, 4, \dots$

Condition for dark rings:

General condition; path difference

$$= (2n + 1)\frac{\lambda}{2}$$

$$\Delta_{eff} = 2\mu t + \frac{\lambda}{2} = (2n + 1)\frac{\lambda}{2}$$

[If the film enclosed has a refractive index μ]

$$2\mu t = n\lambda \quad \dots(3)$$

For air film $\mu=1$

$$\Rightarrow 2t = n\lambda$$

Where $n = 0, 1, 2, 3, 4, \dots$

It is clear from equations (2) and (3) that as the order of fringe increases the thickness of the film increases. The order of fringe at center is zero and it increases as one moves away from center.

Diameter of Rings in Reflected Light

Let 'R' is the radius of curvature of a Plano-convex lens resting on a glass plate PQ. If 't' be the thickness of air film at a point B, a n^{th} order ring of radius $AB = r_n$ is formed there.

From Fig., Using right angle triangle $\triangle OAB$

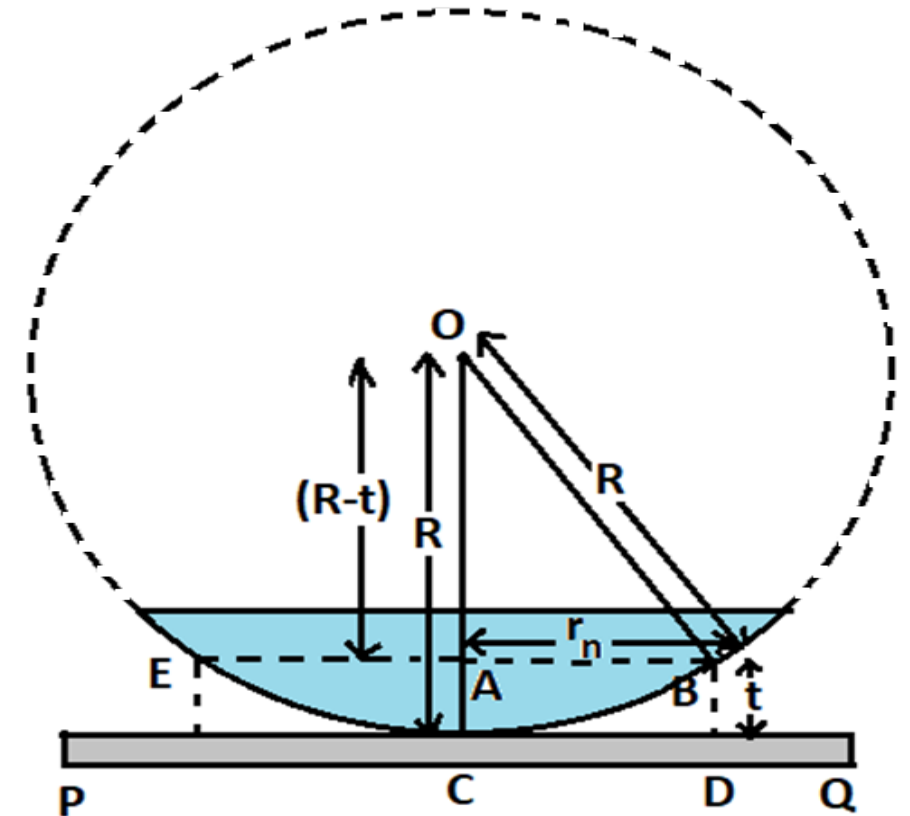
$$(OB)^2 = (OA)^2 + (AB)^2$$

$$R^2 = (R - t)^2 + r_n^2$$

$$R^2 = R^2 + t^2 - 2Rt + r_n^2$$

Since thickness of air film is very small ($t \ll R$) neglecting t^2 term, we have

$$r_n^2 = 2Rt \Rightarrow t = \frac{r_n^2}{2R} \quad \dots(4)$$



Diameter of Dark Ring

From the condition of dark ring

$$2\mu t = n\lambda$$

From equations (4) and (5)

$$\mu \frac{(r_n)_{dark}^2}{R} = n\lambda \Rightarrow (r_n)_{dark}^2 = \frac{Rn\lambda}{\mu}$$

...(5) [where $n=0,1,2,3,\dots$]

[As from equation (4) $2t = \frac{r_n^2}{R}$]

As $(r_n)_{dark} = \frac{(D_n)_{dark}}{2}$ [where $(D_n)_{dark}$ is the diameter of n^{th} dark ring]

$$(D_n)_{dark}^2 = \frac{4Rn\lambda}{\mu} \Rightarrow (D_n)_{dark} = 2\sqrt{\frac{Rn\lambda}{\mu}}$$

...(6)

For air film $\mu = 1$

$$\Rightarrow (D_n)_{dark} = 2\sqrt{Rn\lambda}$$

...(7)

Equations (6) and (7) gives diameters of n^{th} dark rings for film of refractive index μ and air film respectively.

Where $n = 1,2,3,4,\dots$

Diameter of Dark Ring...

It is clear from equation $(D_n)_{dark} = 2\sqrt{\frac{Rn\lambda}{\mu}}$

$$(D_n)_{dark} \propto \sqrt{n}$$

Thus it is clear that, “The **diameter of dark rings is proportional to square root of natural number**”.

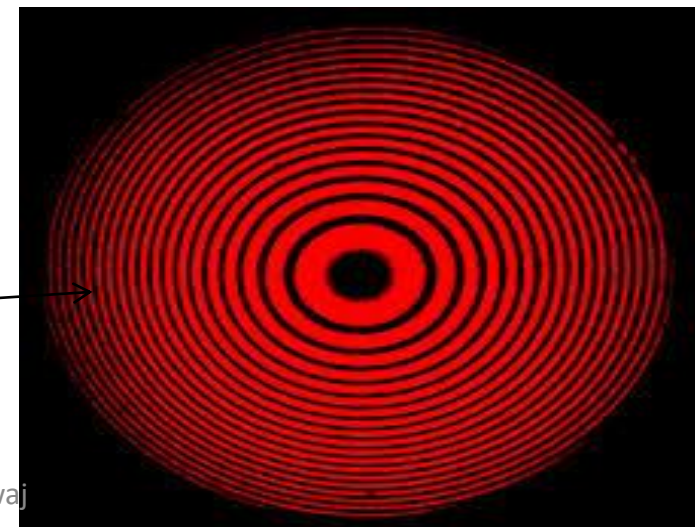
The ratio of diameters of first few dark rings are

$$D_1 : D_2 : D_3 : D_4 : \dots = \sqrt{1} : \sqrt{2} : \sqrt{3} : \sqrt{4} : \dots = 1 : 1.414 : 1.732 : 2 : \dots$$

The separation between successive rings are:

$$\begin{aligned} D_2 - D_1 : D_3 - D_2 : D_4 - D_3 : \dots \\ = 0.414 : 0.318 : 0.268 : \dots \end{aligned}$$

Thus it is clear that dark rings get close as the order increases.



Diameter of Bright Ring

From the condition of bright ring we know that

$$2\mu t = (2n - 1)\frac{\lambda}{2}$$

...(8) [where $n=1,2,3,4,\dots$]

From equations (4) and (8)

$$\mu \frac{(r_n)_{\text{bright}}^2}{R} = (2n - 1)\frac{\lambda}{2}$$

[As from equation (4) $2t = \frac{r_n^2}{R}$]

$$\Rightarrow (r_n)_{\text{bright}}^2 = \frac{(2n - 1)R\lambda}{2\mu}$$

$$\Rightarrow (D_n)_{\text{bright}}^2 = \frac{2(2n - 1)R\lambda}{\mu}$$

[where $(D_n)_{\text{bright}}$ is the diameter of n^{th} bright ring]

$$\Rightarrow (D_n)_{\text{bright}} = \sqrt{\frac{2(2n - 1)R\lambda}{\mu}}$$

...(9) ←

For air film $\mu = 1$

$$\Rightarrow (D_n)_{\text{bright}} = \sqrt{2(2n - 1)R\lambda}$$

...(10) ←

Where $n = 1,2,3,4,\dots$

Equations (9) and (10) gives diameters of n^{th} bright rings for film of refractive index μ and air film respectively.

Diameter of Bright Ring...

It is clear from equation

$$(D_n)_{\text{bright}} = \sqrt{\frac{2(2n-1)R\lambda}{\mu}}$$

$$(D_n)_{\text{bright}} \propto \sqrt{(2n-1)}$$

Thus it is clear that, “The **diameter of bright rings is proportional to square root of odd natural number**”.

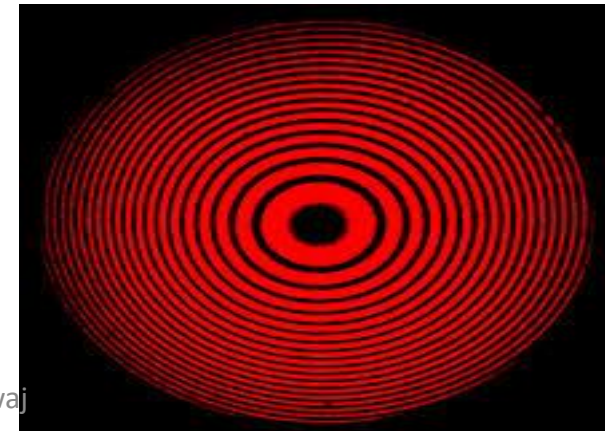
The ratio of diameters first few bright rings are

$$D_1 : D_2 : D_3 : D_4 : \dots = \sqrt{1} : \sqrt{3} : \sqrt{5} : \sqrt{7} : \dots = 1 : 1.732 : 2.236 : 2.646 : \dots$$

The separation between successive rings are:

$$\begin{aligned} D_2 - D_1 : D_3 - D_2 : D_4 - D_3 : \dots \\ = 0.732 : 0.504 : 0.410 : \dots \end{aligned}$$

Thus it is clear that bright rings also get close as the order increases.



Newton's Ring in Transmitted Light

As the ray T_1 is transmitted without any phase change and ray T_2 reflected twice from the optically dense medium (from points C and D) with phase change of 2π or path change of λ and addition and subtraction of λ will make no difference. So, the effective path difference between the transmitted rays will be

$$\Delta_{eff} = 2\mu t \cos(r + \theta) + \frac{\lambda}{2} + \frac{\lambda}{2}$$

$$\Delta_{eff} = 2\mu t \cos(r + \theta) + \lambda$$

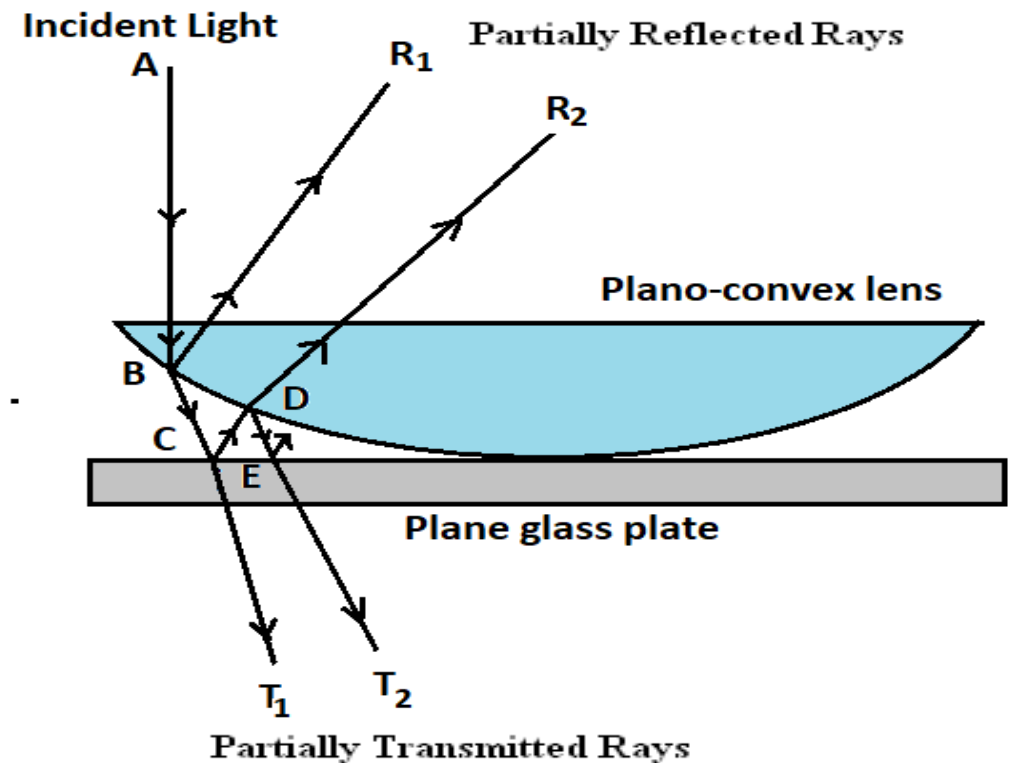
$$\Delta_{eff} = 2\mu t \cos(r + \theta)$$

As radius of curvature of plano convex lens is very large so, neglect θ . For normal incidence $r = 0 \Rightarrow \cos r = 1$ so, the path difference

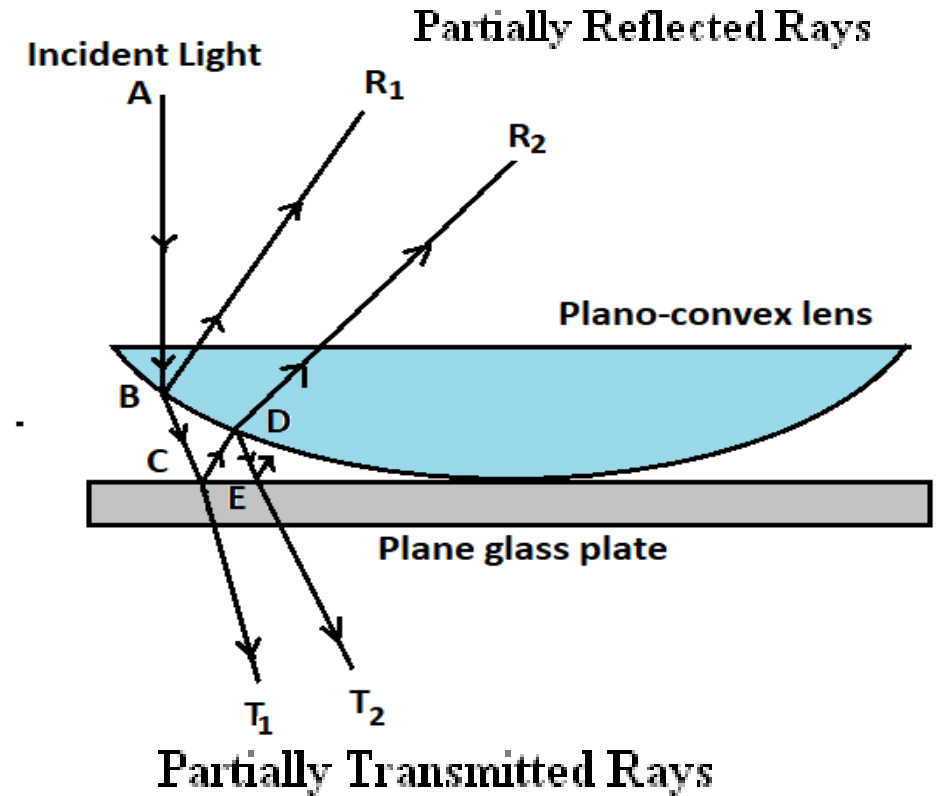
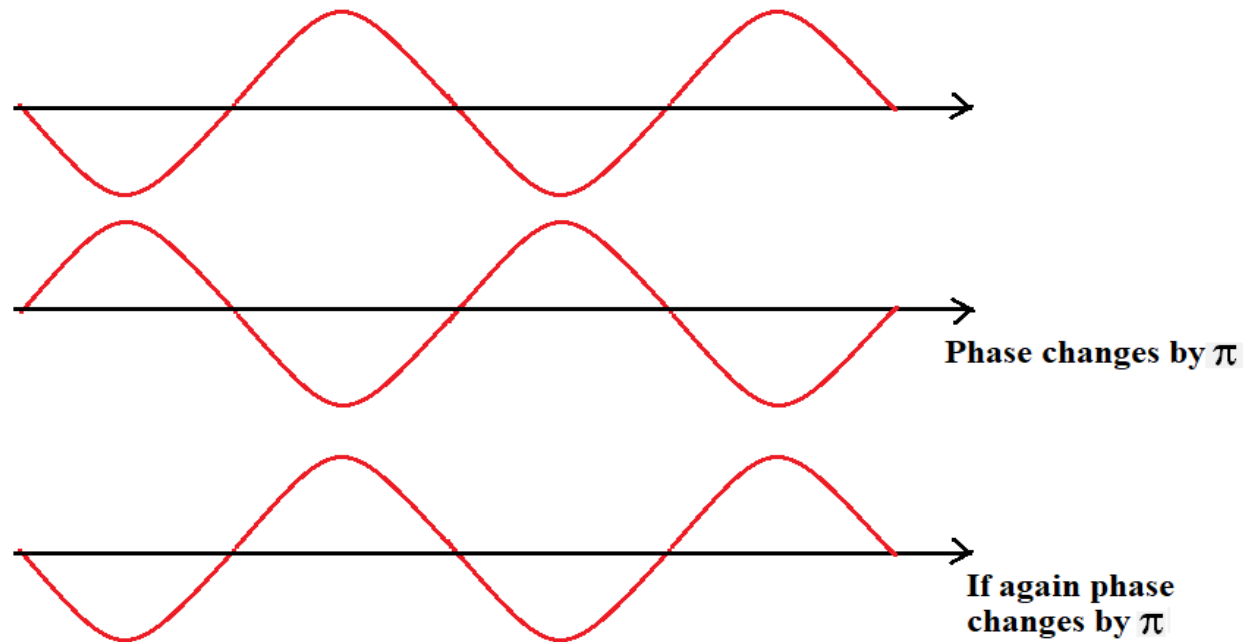
$$\Delta_{eff} = 2\mu t$$

...(11)

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Newton's Ring in Transmitted Light



Newton's Ring in Transmitted Light....

Condition for Dark Ring:

General condition; path difference

$$= (2n - 1) \frac{\lambda}{2}$$

The condition for dark rings

$$2\mu t = (2n - 1) \frac{\lambda}{2} \quad \dots(12)$$

Where $n=1,2,3,4,\dots$

Condition for Bright Ring: General condition;

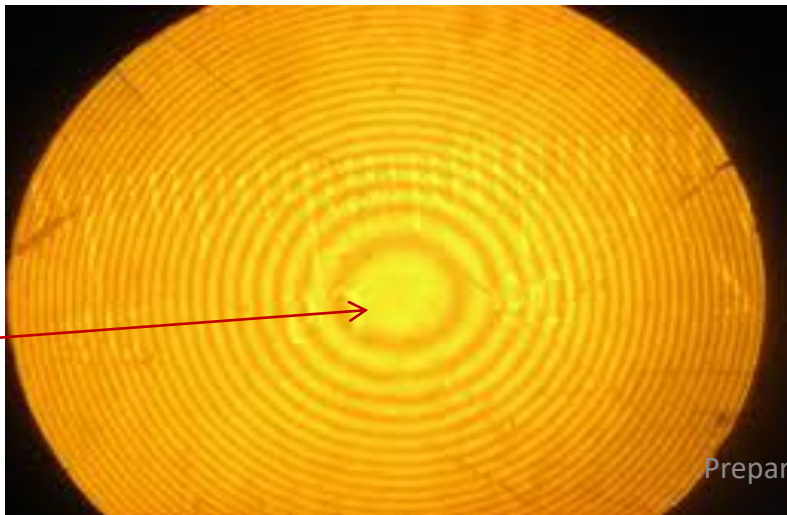
path difference = $n\lambda$

The condition for bright rings

$$2\mu t = n\lambda \quad \dots(13)$$

Where $n=0,1,2,3,\dots$

Bright
centre



Central Spot: At the point of contact between the lens and the plate, $t = 0$,

$\Delta_{eff} = 2\mu t = 0$ **condition for constructive interference.**

So, the central fringe is always bright for transmitted light.

Diameter of Rings in Transmitted Light

Diameter of Dark ring

From equations $2t = \frac{r_n^2}{R}$ & $2\mu t = (2n-1)\frac{\lambda}{2}$

$$\mu \frac{(r_n)_{dark}^2}{R} = (2n-1)\frac{\lambda}{2}$$

$$\Rightarrow (r_n)_{dark}^2 = \frac{(2n-1)R\lambda}{2\mu}$$

$$(r_n)_{dark} = \frac{(D_n)_{dark}}{2}$$

As $\Rightarrow (D_n)_{dark}^2 = \frac{2(2n-1)R\lambda}{\mu}$

$$\Rightarrow (D_n)_{dark} = \sqrt{\frac{2(2n-1)R\lambda}{\mu}} \dots(14)$$

For air film $\mu = 1$

$$\Rightarrow (D_n)_{dark} = \sqrt{2(2n-1)R\lambda} \dots(15)$$

Where $n = 1, 2, 3, 4, \dots$

Diameter of Bright Ring

From equations $2t = \frac{r_n^2}{R}$ & $2\mu t = n\lambda$

$$\mu \frac{(r_n)_{bright}^2}{R} = n\lambda$$

$$\Rightarrow (r_n)_{bright}^2 = \frac{Rn\lambda}{\mu}$$

$$(r_n)_{bright} = \frac{(D_n)_{bright}}{2} \quad \text{As}$$

$$\Rightarrow (D_n)_{bright}^2 = \frac{4n\lambda R}{\mu}$$

$$\Rightarrow (D_n)_{bright} = \sqrt{\frac{4n\lambda R}{\mu}} \dots(16)$$

For air film $\mu = 1$

$$\Rightarrow (D_n)_{bright} = \sqrt{4n\lambda R} \dots(17)$$

Where $n = 1, 2, 3, 4, \dots$

Diameter of Newton's rings in Reflected light:

Diameter of n^{th} dark ring:

$$(D_n)_{\text{dark}} = \sqrt{\frac{4n\lambda R}{\mu}}$$

where $n=1,2,3,\dots$

Diameter of n^{th} bright ring:

$$(D_n)_{\text{bright}} = \sqrt{\frac{2(2n-1)R\lambda}{\mu}}$$

where $n=1,2,3,4,\dots$

Diameter of Newton's rings in Transmitted light:

Diameter of n^{th} dark ring:

$$(D_n)_{\text{dark}} = \sqrt{\frac{2(2n-1)R\lambda}{\mu}}$$

where $n=1,2,3,4,\dots$

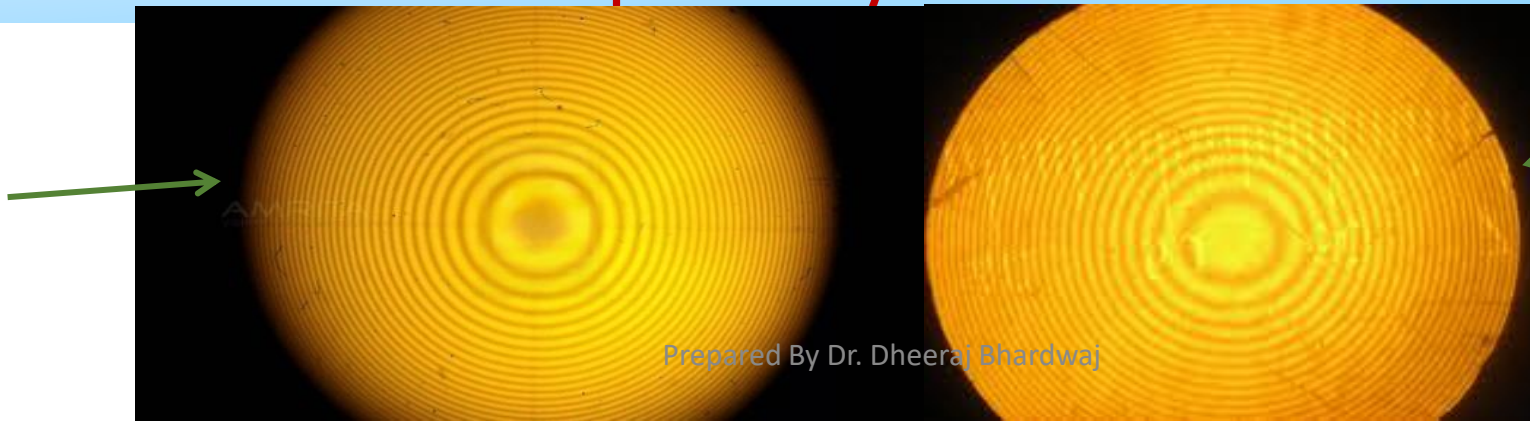
Diameter of n^{th} bright ring

$$(D_n)_{\text{bright}} = \sqrt{\frac{4n\lambda R}{\mu}}$$

where $n=1,2,3,\dots$

In Newton's ring experiment interference fringe pattern in reflected and in transmitted light are complementary to each other.

**Newton's rings
in Reflected
light**



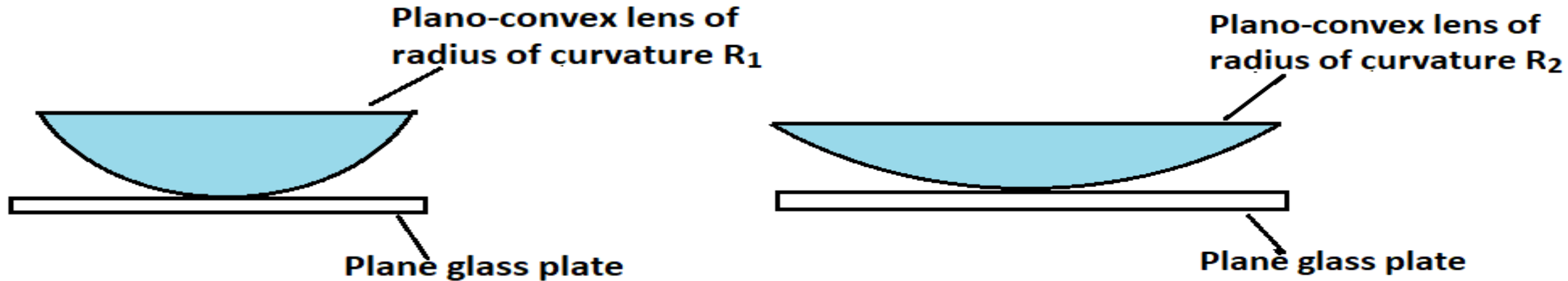
**Newton's rings
in Transmitted
light**

Special Cases

- What will happen in case of Newton's ring experiment if we use a Plano convex lens of small radius of curvature?

As we know that $D_n \propto \sqrt{R}$. Hence if we use a plano-convex lens of small radius of curvature, the diameter of the rings will be small and rings will get closer. This will effect on the percentage error in the measurement of diameter.

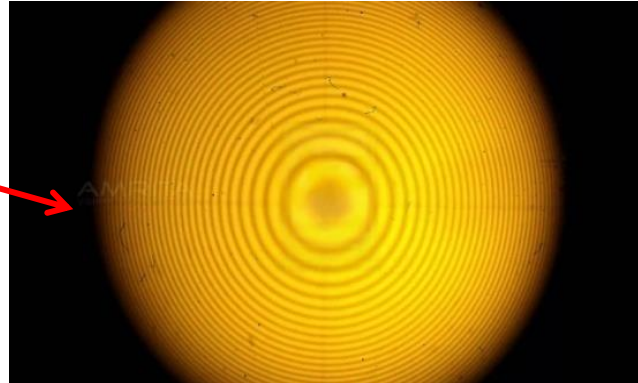
Here $R_2 > R_1$



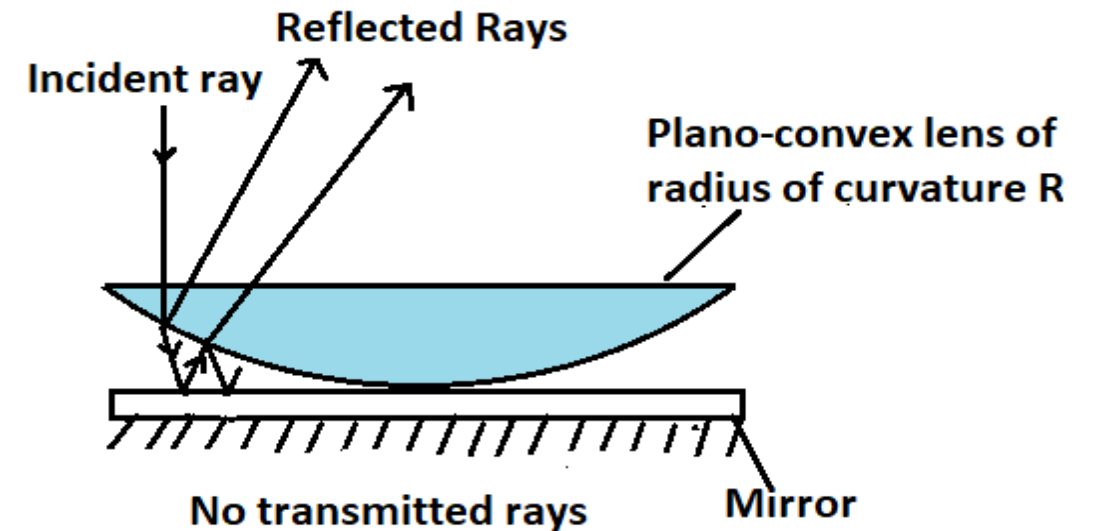
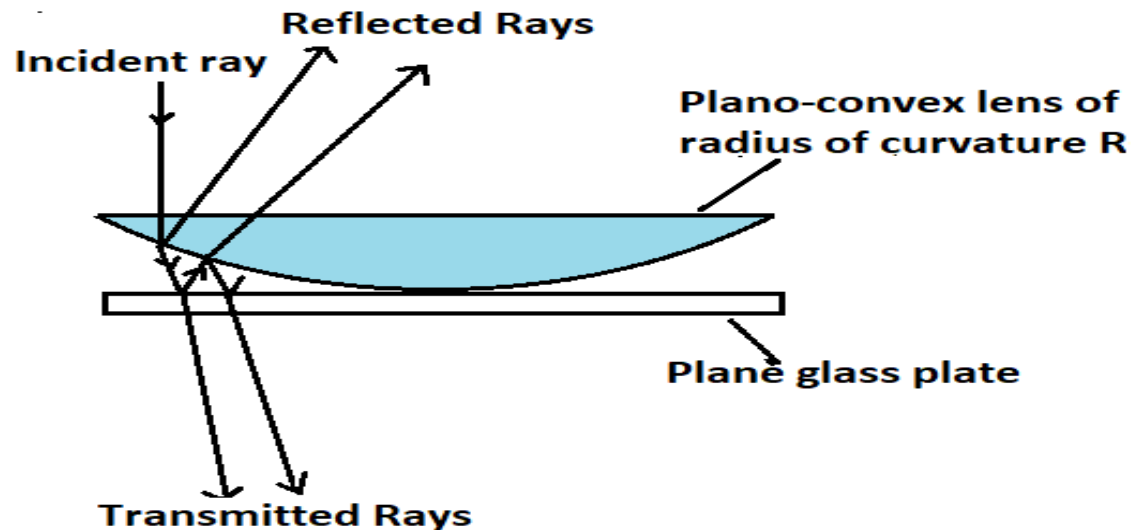
Special Cases.....

- What will happen in case of Newton's ring experiment if we use plane mirror instead of glass plate below the plano-convex lens?

Newton's rings



General Illumination/No fringe pattern



Special Cases.....

- What happens to Newton's ring if the plano-convex lens is raised by some height from the surface of plane glass plate?

Let plano-convex lens is raised by a height Δh from glass plate.

From $\triangle OAB$ $(OA)^2 = (OB)^2 + (AB)^2$

$$R^2 = [R - (t - \Delta h)]^2 + r_n'^2$$

$$R^2 = R^2 + (t - \Delta h)^2 - 2R(t - \Delta h) + r_n'^2$$

$$\Rightarrow r_n'^2 = 2R(t - \Delta h) - (t - \Delta h)^2 \quad \dots(1)$$

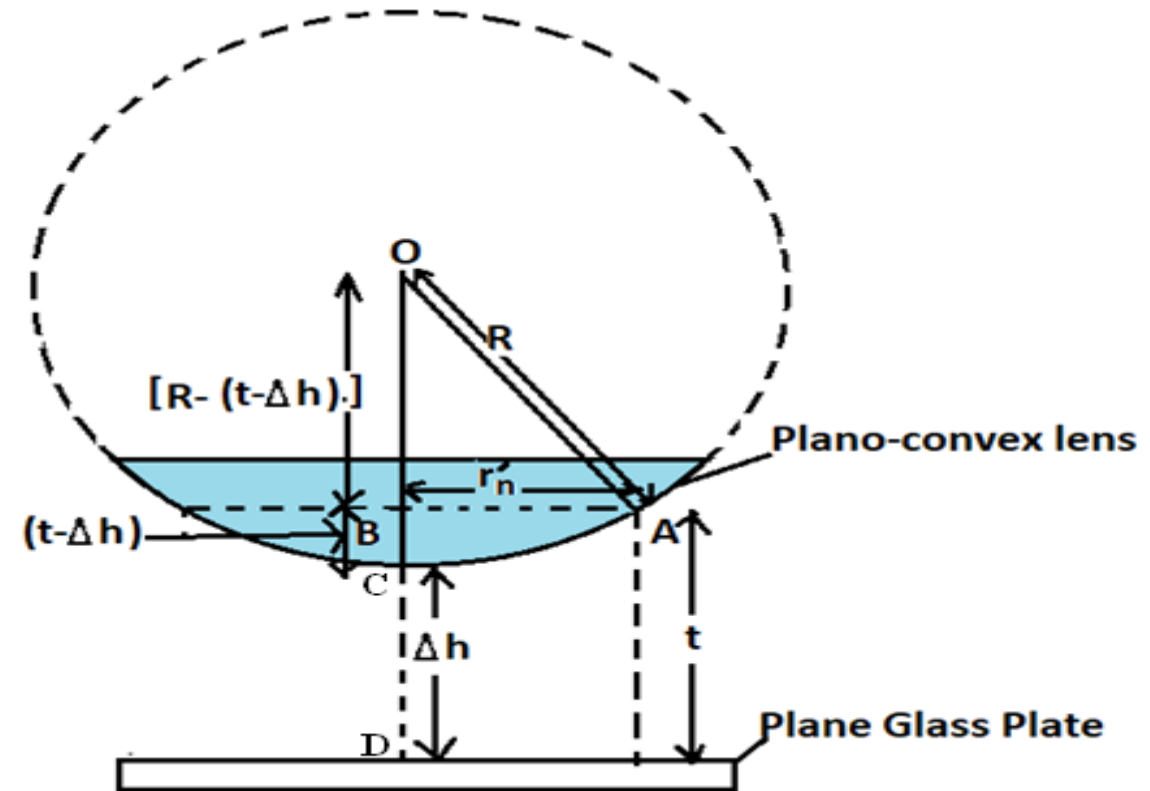
Since the thickness t and shift Δh both are very less, thus neglecting $(t - \Delta h)^2$ compared to $2R(t - \Delta h)$, we get

$$\Rightarrow r_n'^2 = 2Rt - 2R\Delta h$$

Since $r_n^2 = 2Rt$

$$\Rightarrow r_n'^2 = r_n^2 - 2R\Delta h$$

Where r_n is the radius of same n^{th} ring when $\Delta h = 0$



Special Cases.....

For dark ring in reflected light:-

If there is an air film between lens and plate, then

$$(r_n)_{dark}^2 = n\lambda R$$
$$\Rightarrow (r'_n)_{dark}^2 = n\lambda R - 2R\Delta h$$

Thus the radius of the dark ring will be

$$(r'_n)_{dark} = \sqrt{n\lambda R - 2R\Delta h} \quad \dots(2)$$

Similarly for bright ring in reflected light:-

$$(r_n)_{bright}^2 = \frac{(2n-1)\lambda R}{2}$$
$$\Rightarrow (r'_n)_{bright}^2 = (2n-1)\frac{\lambda}{2} R - 2R\Delta h$$

Thus the radius of the bright ring will be

$$(r'_n)_{bright} = \sqrt{(2n-1)\frac{\lambda}{2} R - 2R\Delta h} \quad \dots(3)$$

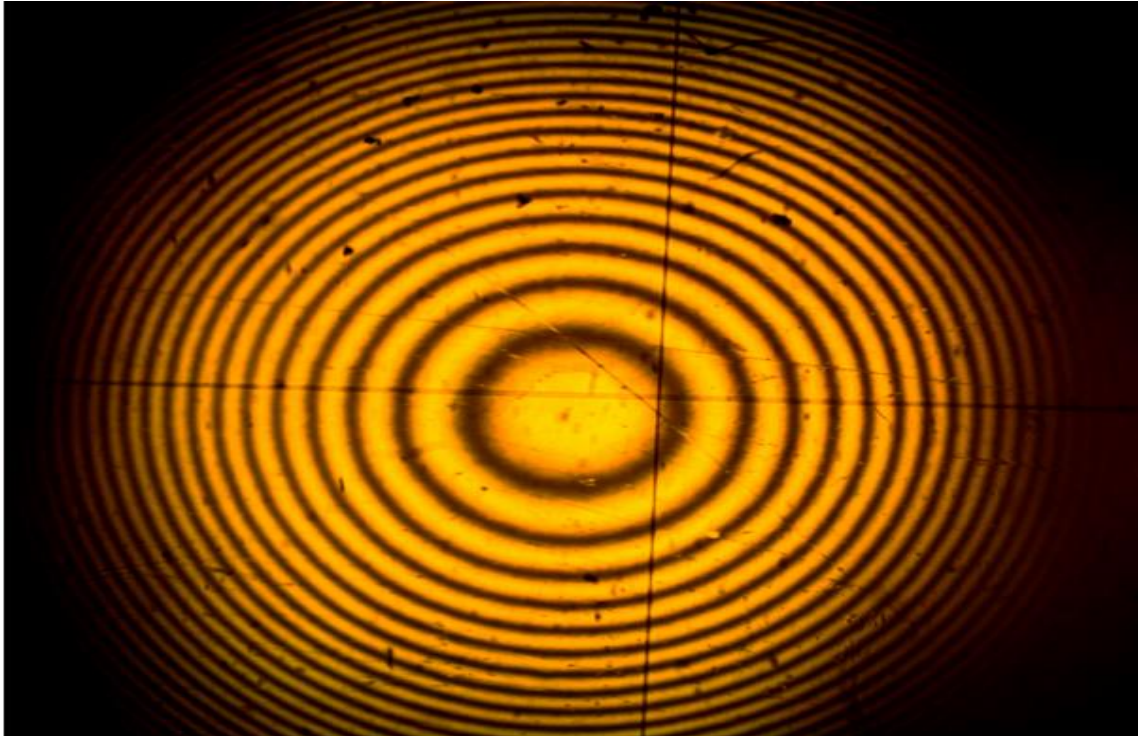
Thus, it is clear from equation (2) and (3) that the radii of Newton's rings decrease on raising the plano-convex lens from glass plate.

Special Cases.....

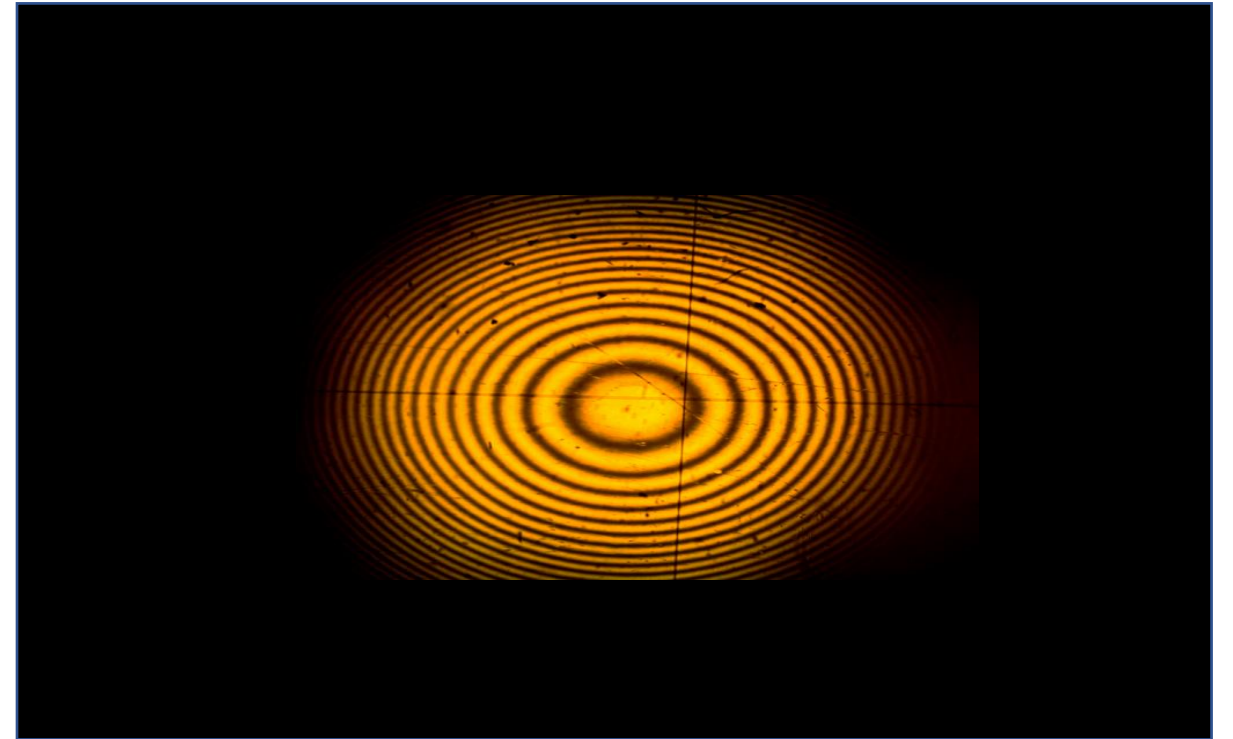
The radius of the bright ring

$$(r'_n)_{\text{bright}} = \sqrt{(2n-1)\frac{\lambda}{2}R - 2R\Delta h}$$

Where r_n is the radius of same n^{th} ring when $\Delta h=0$



When plano convex lens touches
plane glass plate



When plano convex lens is raised by height Δh

Special Cases.....

- What happens in case of Newton's ring if a plane glass plate below the plano-convex lens is replaced by another plano-convex lens?

Let the radius of curvatures of two plano-convex lenses be R_1 and R_2 respectively. Where $R_1 > R_2$ and they are touching each other. A thin air film is enclosed in between two curved surfaces.

Let the thickness of thin film at an arbitrary point P is t . Suppose n^{th} order ring is formed here, whose radius is r_n .

Here $t = t_1 + t_2$... (4)

From geometry we can write

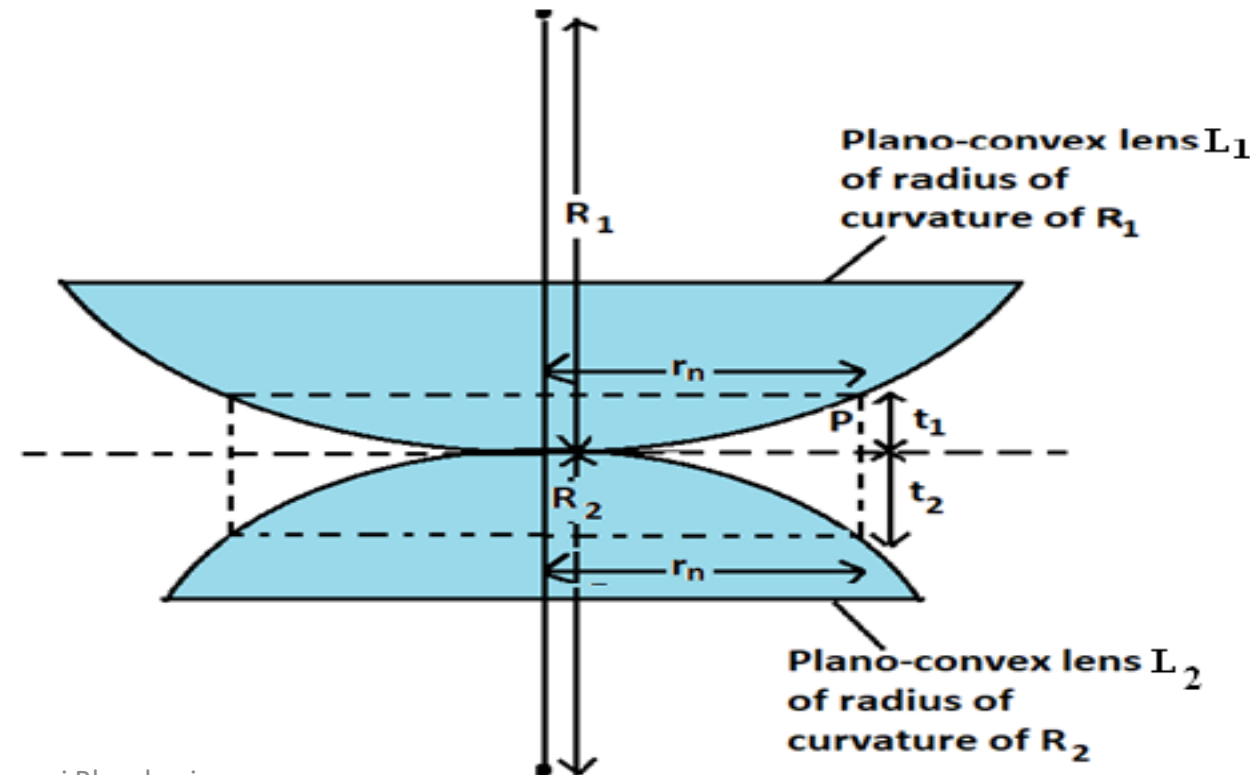
$$r_n^2 = 2R_1 t_1 = 2R_2 t_2$$

$$\Rightarrow t_1 = \frac{r_n^2}{2R_1} \text{ and } t_2 = \frac{r_n^2}{2R_2}$$

$$\Rightarrow t = \frac{r_n^2}{2R_1} + \frac{r_n^2}{2R_2}$$

From equation (4)

$$\Rightarrow t = \frac{r_n^2}{2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \quad \dots (5)$$



For dark rings in reflected light:

$$2\mu t = n\lambda \quad \dots(6)$$

From equations (5) and (6), we get

$$2\mu \frac{r_n^2}{2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = n\lambda \quad \dots(7)$$

$$\Rightarrow r_n^2 = \frac{n\lambda R_1 R_2}{\mu(R_2 + R_1)}$$

$$\Rightarrow (r_n)_{dark} = \sqrt{\frac{n\lambda R_1 R_2}{\mu(R_2 + R_1)}}$$

$$D_n = 2r_n$$

As $\Rightarrow (D_n)_{dark} = \sqrt{\frac{4n\lambda R_1 R_2}{\mu(R_2 + R_1)}}$

For air film

$$\mu = 1$$

$$\Rightarrow [(D_n)_{dark}]_{air} = \sqrt{\frac{4n\lambda R_1 R_2}{(R_2 + R_1)}} \quad \dots(8)$$

Where $n=1,2,3,4,\dots$

For bright rings in reflected light:

$$2\mu t = (2n - 1) \frac{\lambda}{2} \quad \dots(9)$$

From equations (5) and (9), we get

$$2\mu \frac{r_n^2}{2} \left(\frac{1}{R_1} + \frac{1}{R_2} \right) = (2n - 1) \frac{\lambda}{2}$$

$$\Rightarrow r_n^2 = \frac{(2n - 1)\lambda R_1 R_2}{2\mu(R_2 + R_1)}$$

$$\Rightarrow (r_n)_{bright} = \sqrt{\frac{(2n - 1)\lambda R_1 R_2}{2\mu(R_2 + R_1)}} \quad \dots(10)$$

As $D_n = 2r_n$

$$\Rightarrow (D_n)_{bright} = \sqrt{\frac{2(2n - 1)\lambda R_1 R_2}{\mu(R_2 + R_1)}}$$

For air film

$$\mu = 1$$

$$\Rightarrow [(D_n)_{bright}]_{air} = \sqrt{\frac{2(2n - 1)\lambda R_1 R_2}{(R_2 + R_1)}} \quad \dots(11)$$

Where $n=1,2,3,\dots$

Special Cases.....

- What happens in case of Newton's ring if plane glass plate below the plano-convex lens is replaced by another plano-concave lens?

Let the thickness of thin film at an arbitrary point P is t . Suppose n^{th} order ring is formed here, whose radius is r_n . Where $R_2 > R_1$

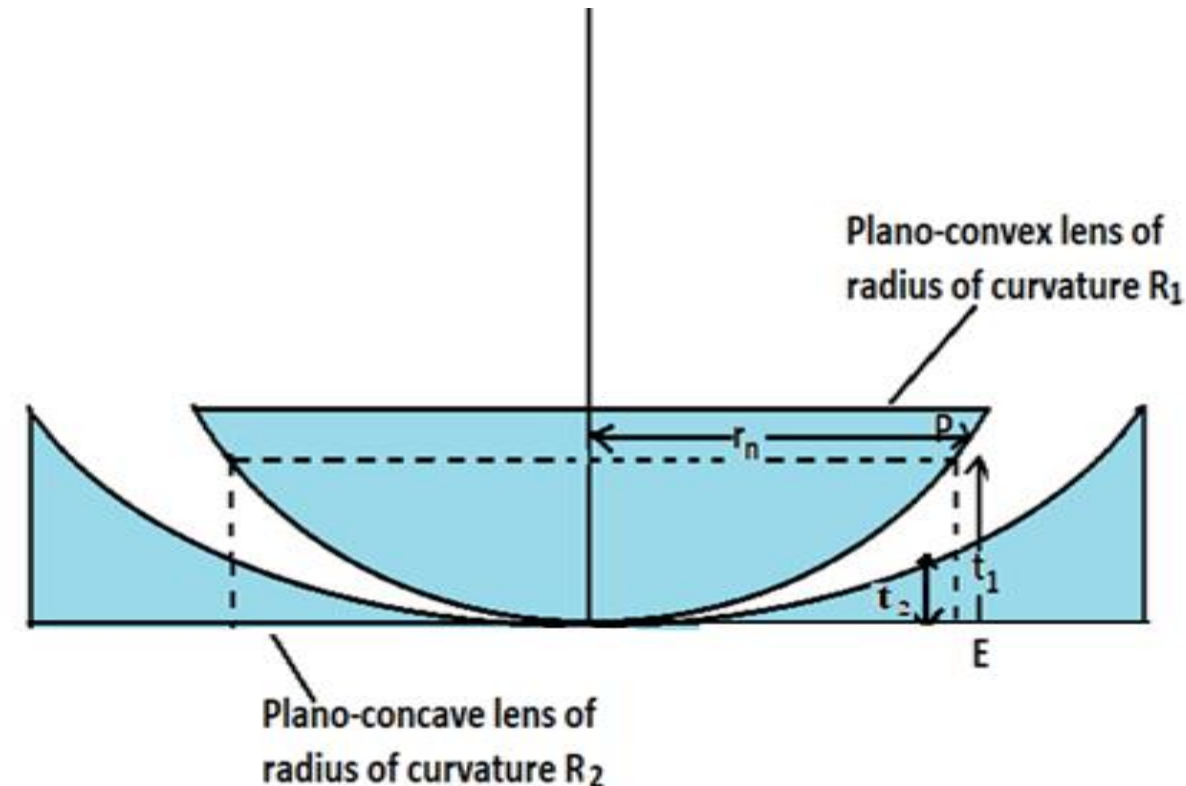
Here $r_n^2 = 2R_1t_1 = 2R_2t_2$... (12)

From geometry we can write

$$\begin{aligned} t &= t_1 - t_2 \\ \Rightarrow t_1 &= \frac{r_n^2}{2R_1} \quad \text{and} \quad t_2 = \frac{r_n^2}{2R_2} \end{aligned}$$

From equation (4)

$$\begin{aligned} \Rightarrow t &= \frac{r_n^2}{2R_1} - \frac{r_n^2}{2R_2} \\ \Rightarrow t &= \frac{r_n^2}{2} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \end{aligned} \quad \dots (13)$$



For dark rings in reflected light:

$$2\mu t = n\lambda \quad \dots(6)$$

From equations (13) and (6), we get

$$\begin{aligned} 2\mu \frac{r_n^2}{2} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) &= n\lambda \\ \Rightarrow r_n^2 &= \frac{n\lambda R_1 R_2}{\mu(R_2 - R_1)} \\ \Rightarrow (r_n)_{dark} &= \sqrt{\frac{n\lambda R_1 R_2}{\mu(R_2 - R_1)}} \quad \dots(14) \end{aligned}$$

As $D_n = 2r_n$

$$\Rightarrow (D_n)_{dark} = \sqrt{\frac{4n\lambda R_1 R_2}{\mu(R_2 - R_1)}}$$

For air film
 $\mu = 1$

$$\Rightarrow [(D_n)_{dark}]_{air} = \sqrt{\frac{4n\lambda R_1 R_2}{(R_2 - R_1)}} \quad \dots(15)$$

Where $n=1,2,3,4,\dots$

For bright rings in reflected light:

$$2\mu t = (2n - 1) \frac{\lambda}{2} \quad \dots(9)$$

From equations (13) and (9), we get

$$\begin{aligned} 2\mu \frac{r_n^2}{2} \left(\frac{1}{R_1} - \frac{1}{R_2} \right) &= (2n - 1) \frac{\lambda}{2} \\ \Rightarrow r_n^2 &= \frac{(2n - 1)\lambda R_1 R_2}{2\mu(R_2 - R_1)} \\ \Rightarrow (r_n)_{bright} &= \sqrt{\frac{(2n - 1)\lambda R_1 R_2}{2\mu(R_2 - R_1)}} \quad \dots(16) \end{aligned}$$

As $D_n = 2r_n$

$$\Rightarrow (D_n)_{bright} = \sqrt{\frac{2(2n - 1)\lambda R_1 R_2}{\mu(R_2 - R_1)}}$$

For air film $\mu = 1$

$$\Rightarrow [(D_n)_{bright}]_{air} = \sqrt{\frac{2(2n - 1)\lambda R_1 R_2}{(R_2 - R_1)}} \quad \dots(17)$$

Where $n=1,2,3,4,\dots$

Special Cases.....

➤ **What happens in case of Newton's ring if plano-convex lens is replaced by plano-concave lens?**

- In this case fringes are still circular.
- As the lens touches the glass plate at its periphery hence dark ring will be observed there.
- As we move away from the periphery to centre the thickness of air film between two surfaces gradually increases hence the higher order ring will be observed at centre where the thickness of air film is highest and zero order ring will be observed at periphery of the lens.
- Let 'H' be the thickness of air film at centre and t_n is the thickness of film for n^{th} order ring, and R be the radius of curvature of plano-concave lens, then radius of n^{th} order ring will be

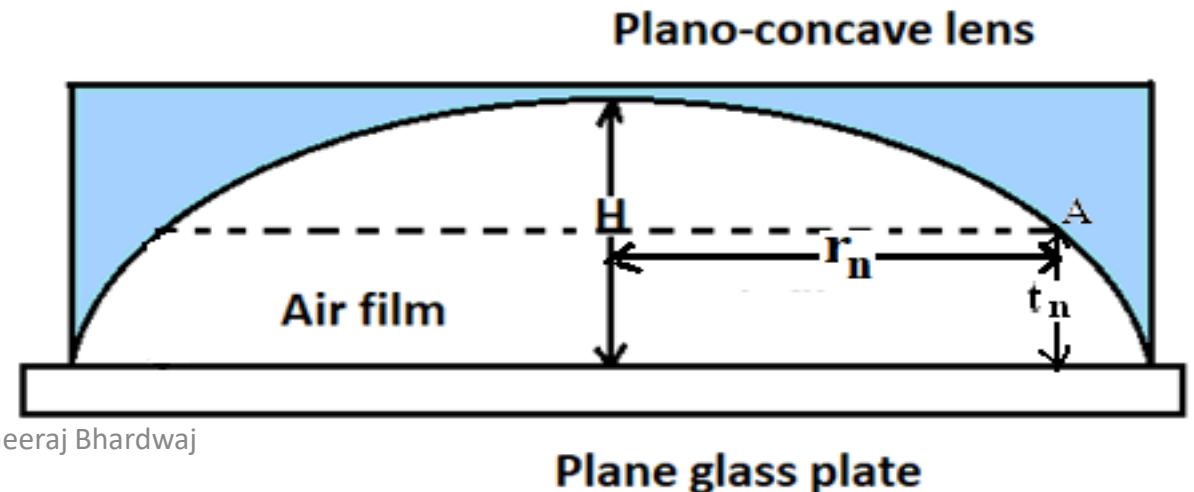
$$r_n = \sqrt{2(H - t_n)R}$$

- For dark ring in reflected light

$$(r_n)_{\text{dark}} = \sqrt{\left(2H - \frac{n\lambda}{\mu}\right)R}$$

- For bright ring in reflected light

$$(r_n)_{\text{bright}} = \sqrt{\left[2H - (2n - 1)\frac{\lambda}{2\mu}\right]R}$$



Special Cases.....

From fig. $OA=OB=R$ = Radius of curvature of plano-concave lens.

Thickness of air film at centre $BE=H$

At point A n^{th} order ring is observed. Let the radius of this ring be r_n .

Thickness of air film at point A $=t_n$.

$$BC=(H-t_n)$$

$$CO=[R-(H-t_n)]$$

From $\triangle AOC$

$$(AO)^2 = (OC)^2 + (CA)^2$$

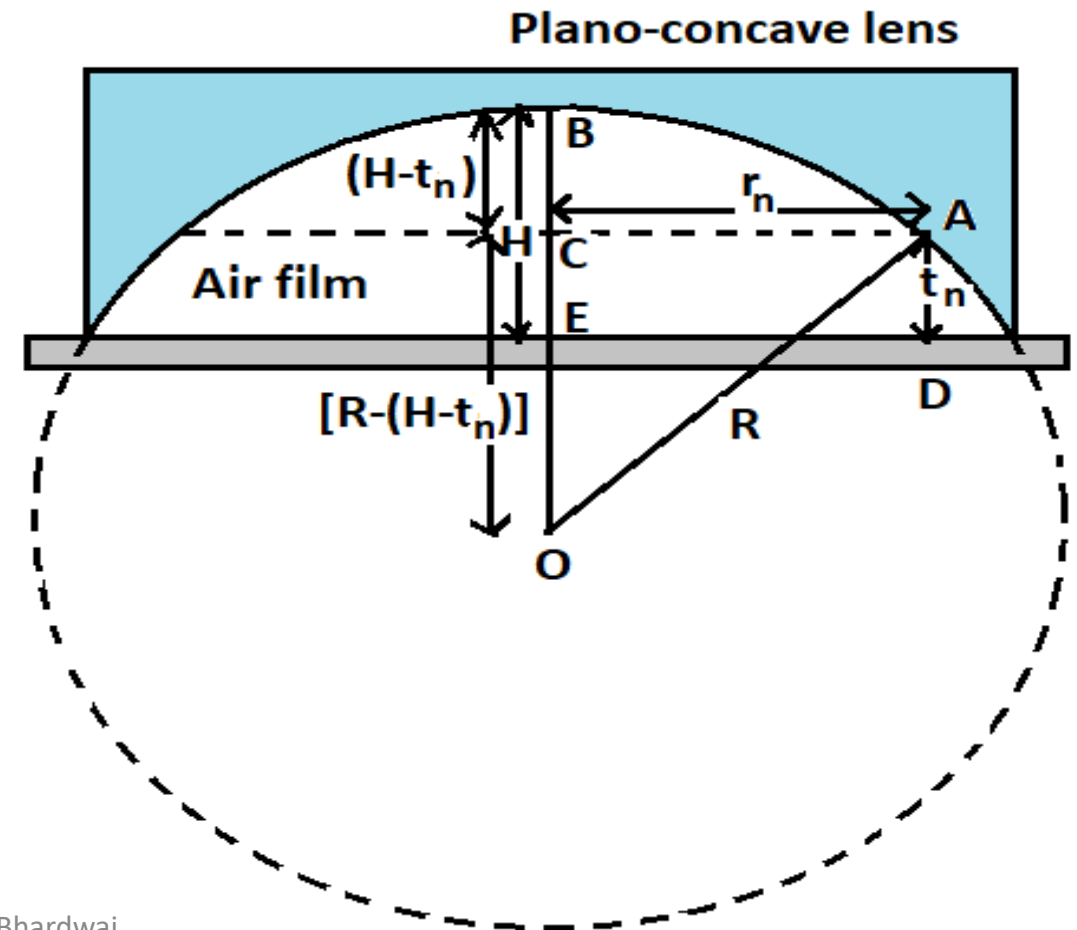
$$R^2 = [R - (H - t_n)]^2 + r_n^2$$

$$R^2 = R^2 + (H - t_n)^2 - 2R(H - t_n) + r_n^2$$

As $(H-t_n) \ll R$ so neglecting $(H-t_n)^2$ as compared to R .

$$r_n^2 = 2R(H - t_n)$$

$$r_n = \sqrt{2R(H - t_n)}$$

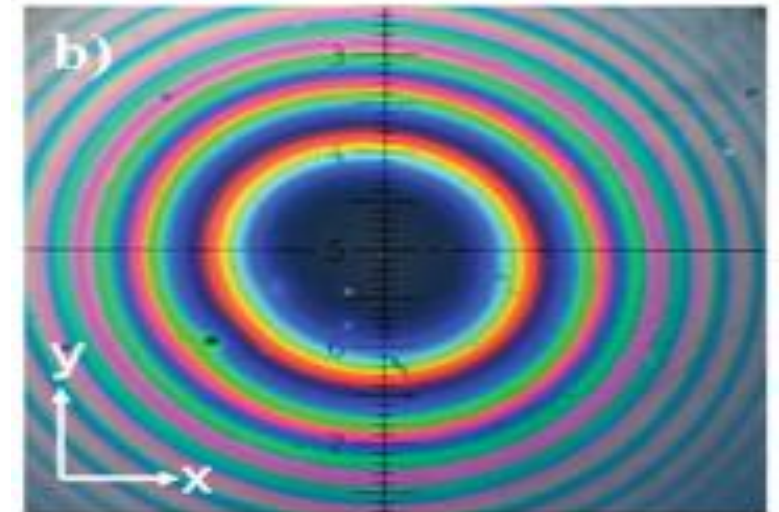
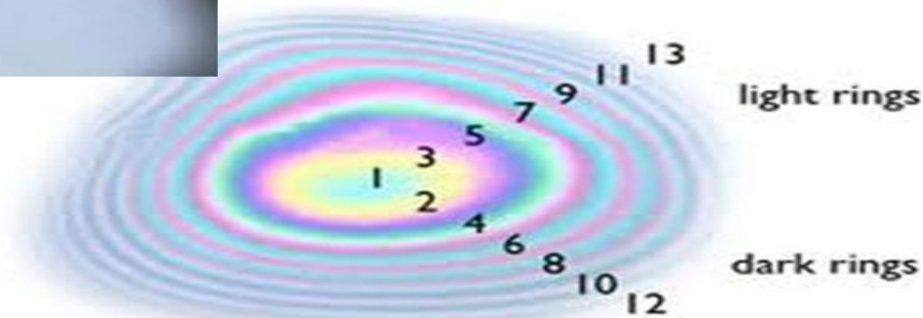
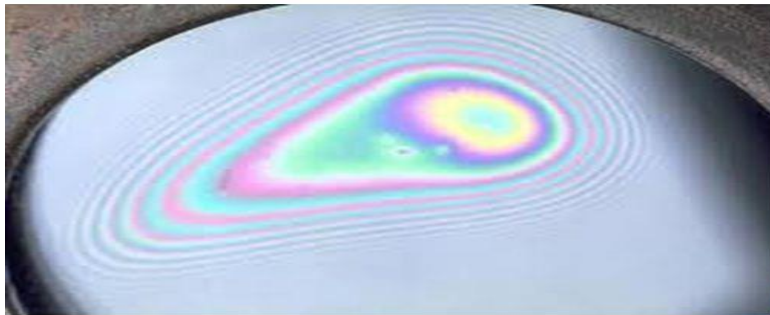


Special Cases.....

➤ Newton's ring with white light:

With monochromatic light in Newton's ring experiment we get alternate bright and dark rings. As we know that $D_n \propto \sqrt{\lambda}$ and white light consist of seven colours having wavelength region from 4000 Å to 8000 Å. Therefore with white light few coloured rings are observed and after that due to overlapping of rings of higher order and different wavelengths a general illumination is obtained and no rings are observed.

Newton Rings... In white light



Special Cases.....

➤ Bright centre in reflected pattern in Newton's ring :

A bright centre in reflected system can be obtained by selecting plano-convex lens, medium (film) and the glass plate in such a way that:

❖ $\mu_1 < \mu < \mu_2$ So that there is phase reversal of π at upper and lower surface of the film. In this case $\Delta_{\text{eff}} = 2\mu t + \lambda/2 + \lambda/2$

i.e $\Delta_{\text{eff}} = 2\mu t + \lambda$

At the point of contact $t = 0$,

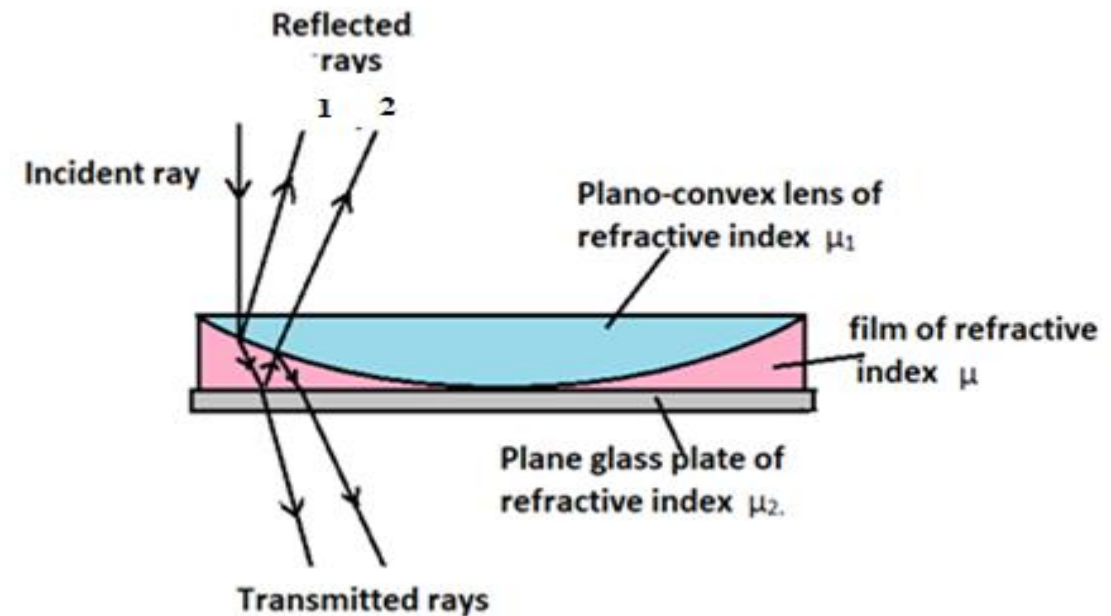
$\Rightarrow \Delta_{\text{eff}} = \lambda \rightarrow$ **condition of constructive interference**

Therefore bright centre can be obtained.

❖ $\mu_1 > \mu > \mu_2$. In this case $\Delta = 2\mu t$ as both rays are reflected from dense to rare boundary and at the point of contact $t = 0$,

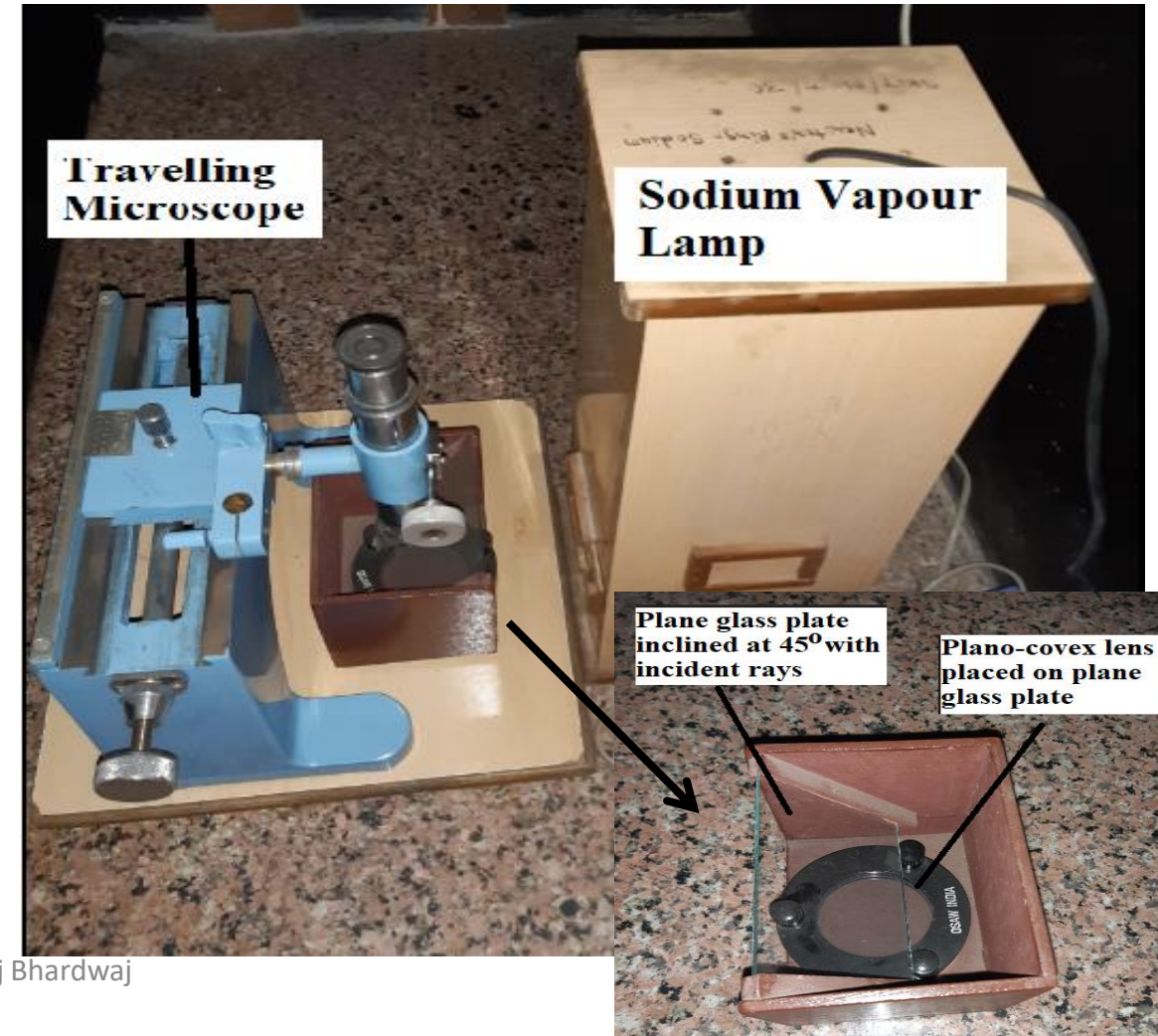
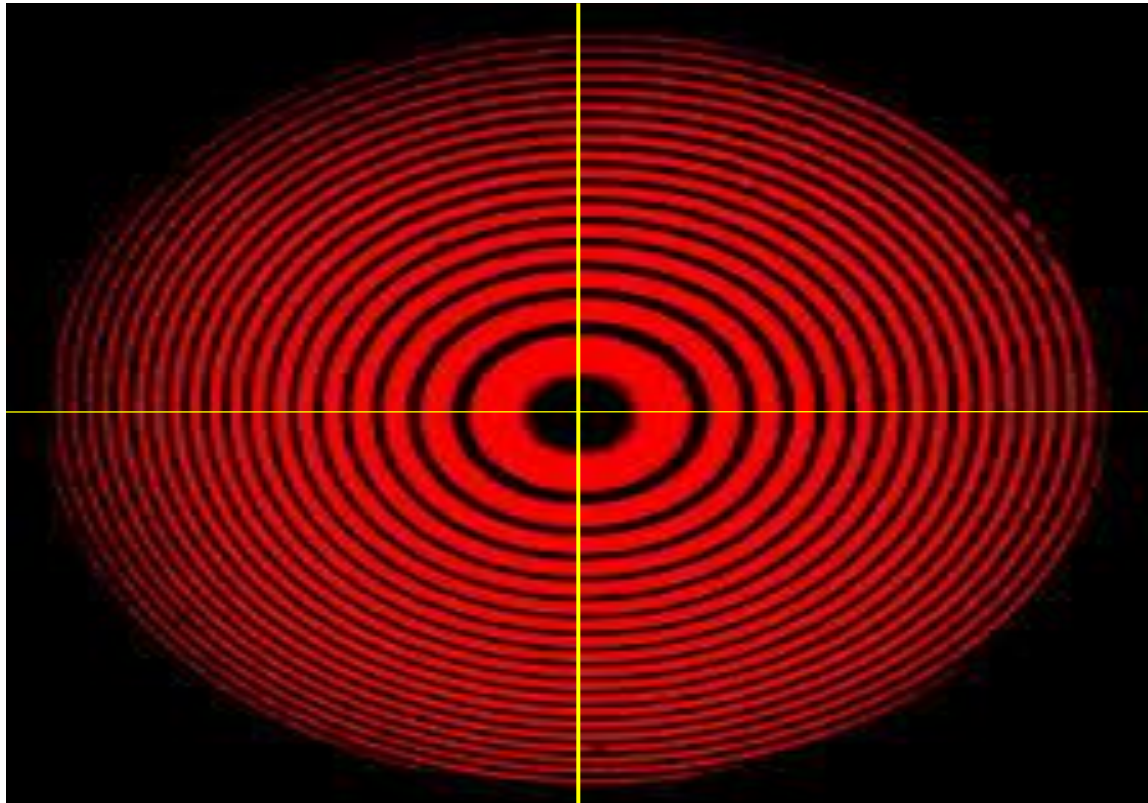
$\Rightarrow \Delta_{\text{eff}} = 0 \rightarrow$ **condition of constructive interference**

➡ **Bright centre**



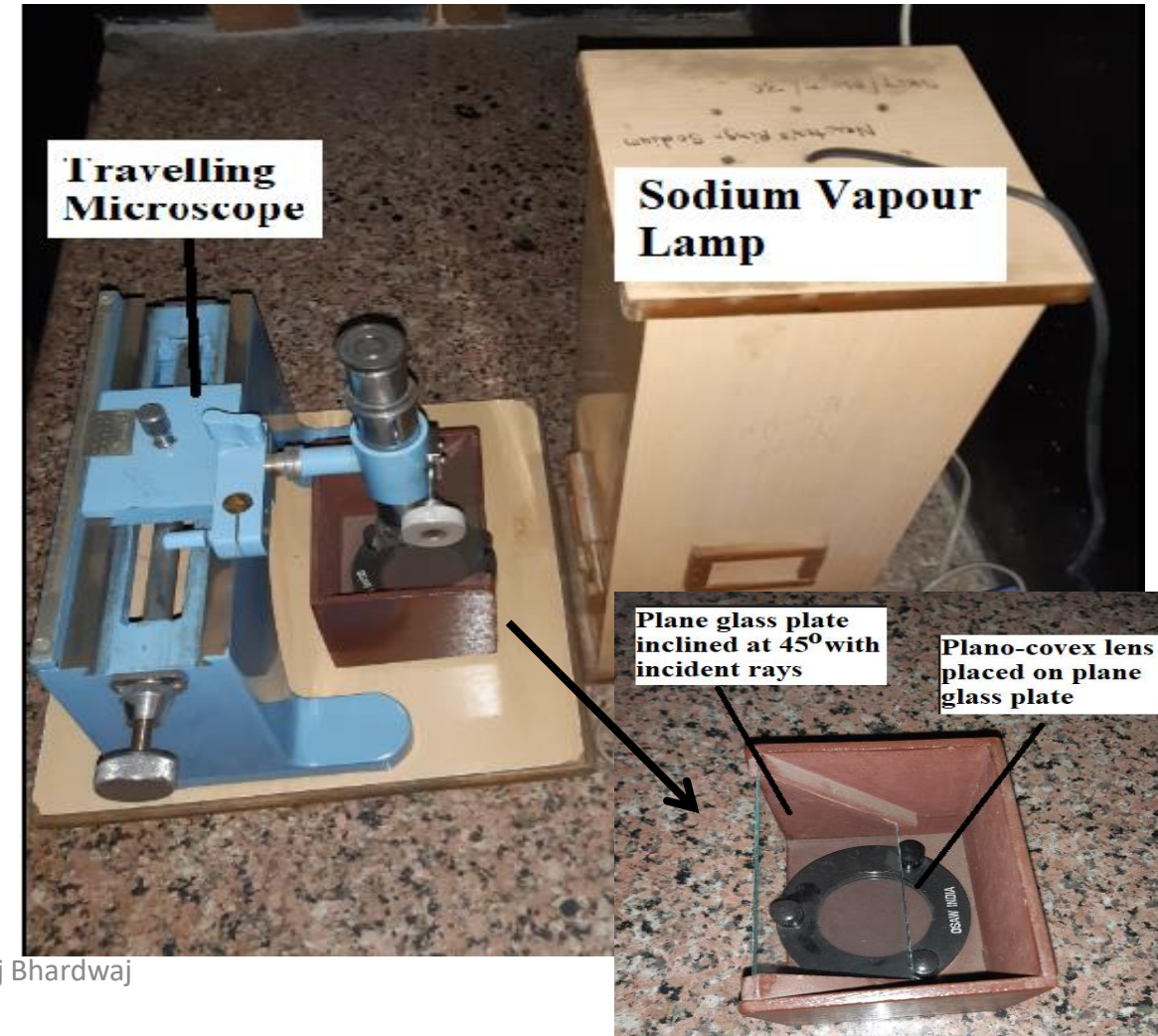
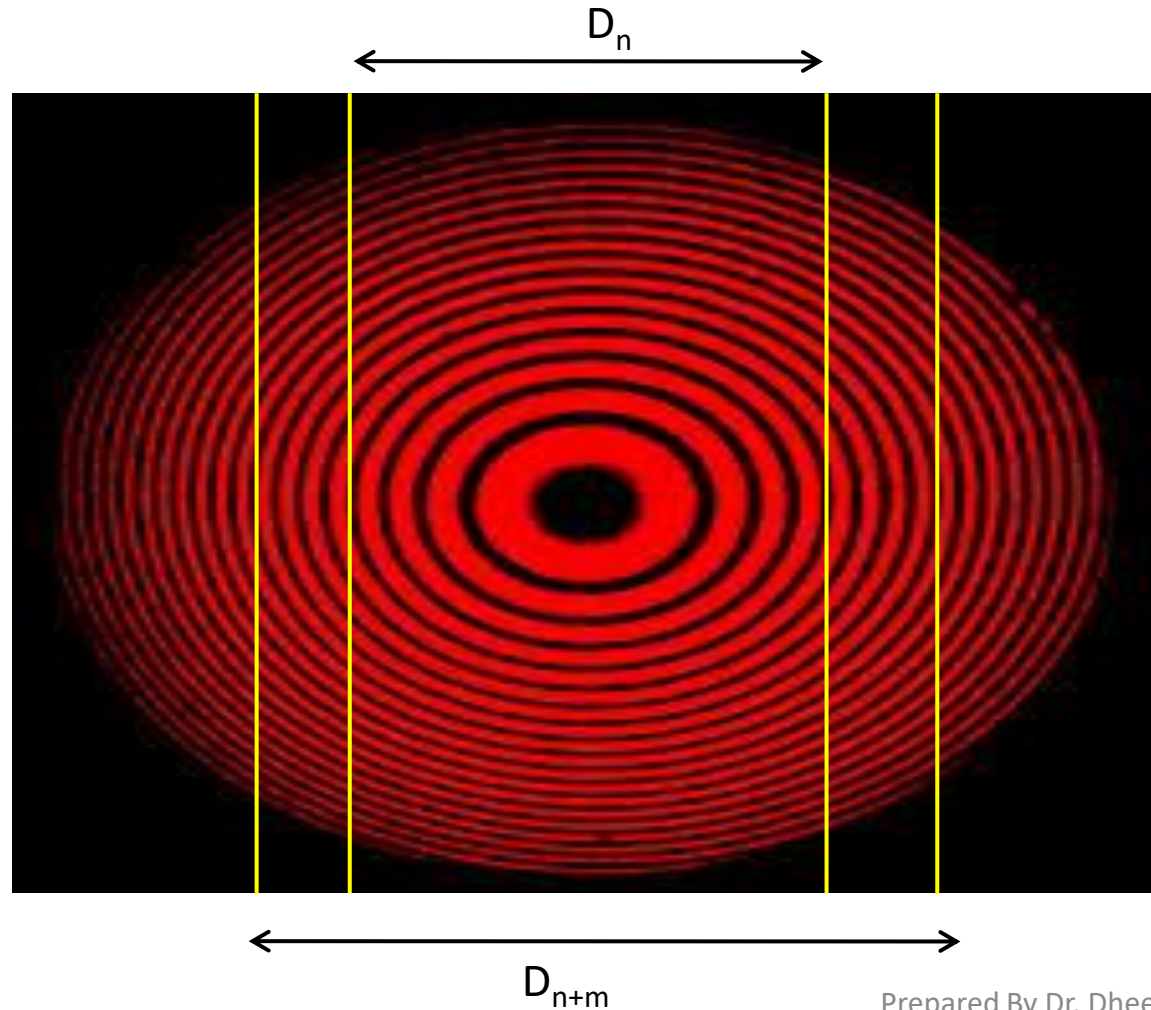
Applications of Newton's Ring Experiment

Determination of Wavelength of Monochromatic Light



Applications of Newton's Ring Experiment

Determination of Wavelength of Monochromatic Light



Applications of Newton's Ring Experiment.....

- The diameter of the n^{th} order dark ring is calculated by subtracting the left and right side position of the microscope for n^{th} order dark ring . As we know that the square of diameter of n^{th} dark ring is

$$D_n^2 = \frac{4n\lambda R}{\mu} \quad \dots(1)$$

For air film $\mu = 1$

$$\Rightarrow D_n^2 = 4n\lambda R \quad \dots(2)$$

- In the same way find the diameter of of $(n+m)^{\text{th}}$ dark ring and the square of diameter of $(n+m)^{\text{th}}$ ring (for air film) is

$$D_{(n+m)}^2 = 4(n+m)\lambda R \quad \dots(3)$$

- Taking difference between the equations (2)

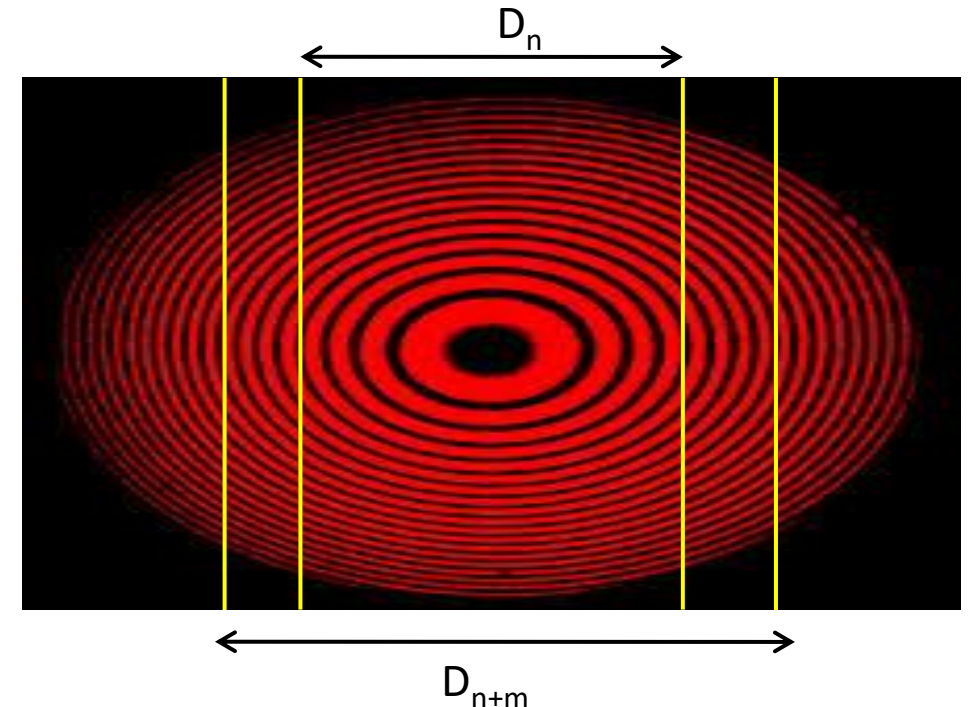
and (3)

$$D_{(n+m)}^2 - D_n^2 = 4(n+m)\lambda R - 4n\lambda R$$

$$\Rightarrow D_{(n+m)}^2 - D_n^2 = 4m\lambda R$$

$$\Rightarrow \lambda = \frac{D_{(n+m)}^2 - D_n^2}{4mR} \quad \dots(4)$$

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Applications of Newton's Ring Experiment.....

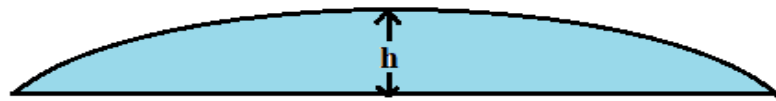
- Here R is the radius of curvature of plano convex lens and can be measured by using a spherometer, using the formula

$$R = \frac{a^2}{6h} + \frac{h}{2}$$

Where

a = separation between two legs of spherometer

h=height (depth) of convexite of lens



Plano-convex lens

Graphical Method

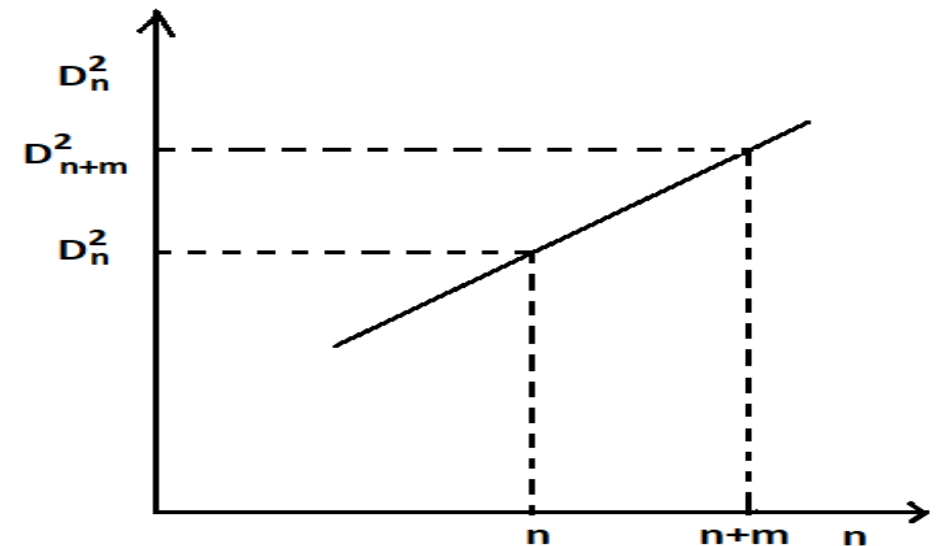
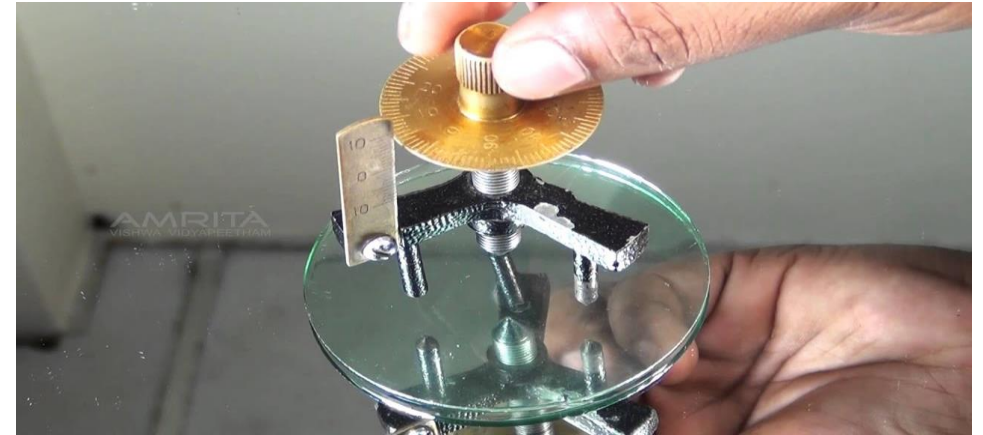
$$\text{Slope of the line } D_n^2 \text{ Vs } n = \frac{D_{n+m}^2 - D_n^2}{m}$$

So wavelength of monochromatic light can be determined as

$$\lambda = \frac{\text{slope of line } D_n^2 \text{ Vs } n}{4R}$$

...(5)

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Applications of Newton's Ring Experiment.....

Determination of refractive index of liquid in small quantity

- In order to determine the refractive index of liquid the Newton's ring experiment is performed for the air film and the square of diameters of $(n+m)^{\text{th}}$ and n^{th} dark ring are determined.

- Square of diameter of $(n+m)^{\text{th}}$ dark ring in reflected light (for air film)

$$[D_{(n+m)}^2]_{\text{air}} = 4(n+m)\lambda R \quad \dots(6)$$

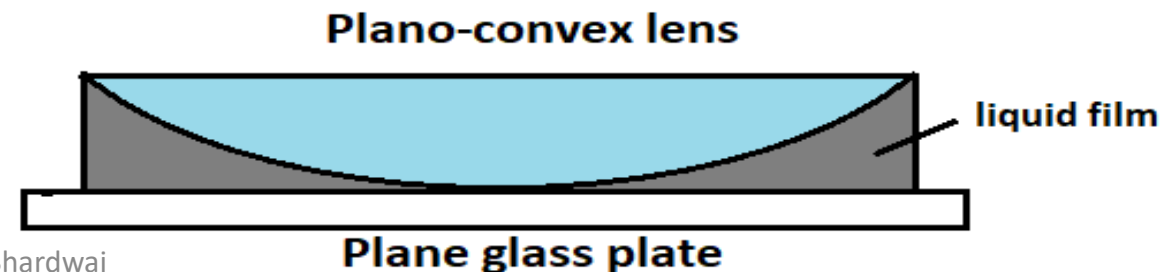
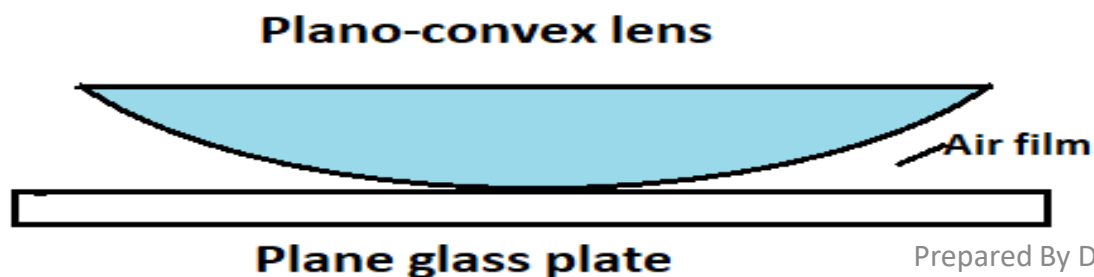
- Square of diameter of n^{th} dark ring in reflected light (for air film)

$$[D_n^2]_{\text{air}} = 4n\lambda R \quad \dots(7)$$

- Taking difference between both the equations

$$[D_{(n+m)}^2 - D_n^2]_{\text{air}} = 4m\lambda R \quad \dots(8)$$

- After this a liquid of refractive index μ is inserted between plano convex lens and glass plate and the square of diameters of same $(n+m)^{\text{th}}$ and n^{th} dark ring are determined.



Applications of Newton's Ring Experiment.....

- Square of diameter of $(n+m)^{\text{th}}$ dark ring in reflected light (for liquid of refractive index μ)

$$[D_{(n+m)}^2]_{\text{liquid}} = \frac{4(n+m)\lambda R}{\mu} \quad \dots(9)$$

- Square of diameter of n^{th} dark ring in reflected light (for liquid of refractive index μ)

$$[D_n^2]_{\text{liquid}} = \frac{4n\lambda R}{\mu} \quad \dots(10)$$

- Taking difference between both the equations

$$\mu[D_{(n+m)}^2 - D_n^2]_{\text{liquid}} = 4m\lambda R \quad \dots(11)$$

- On dividing equation (8) by equation (11)

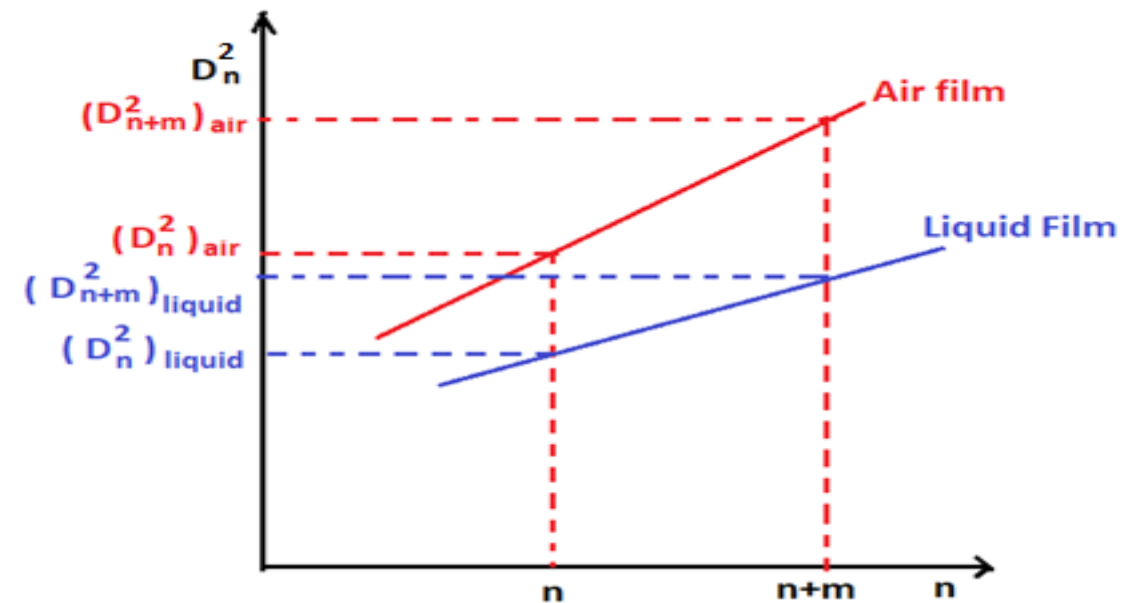
$$\mu = \frac{[D_{(n+m)}^2 - D_n^2]_{\text{air}}}{[D_{(n+m)}^2 - D_n^2]_{\text{liquid}}} \quad \dots(12)$$

Hence refractive index can be determined by measuring the diameter of Newton's rings first with air and with the liquid for same value of m .

Applications of Newton's Ring Experiment.....

Graphical method to determine refractive index of liquid in small quantity

$$\mu = \frac{\text{Slope of line } D_n^2 \text{ Vs } n \text{ for air film}}{\text{Slope of line } D_n^2 \text{ Vs } n \text{ for liquid film}}$$



Summary

- Newton's rings are concentric dark and bright rings observed when film enclosed between plano-convex lens and plane glass plate is illuminated by monochromatic light.
- In Newton's ring experiment two coherent light rays are produced by division of amplitude.
- These fringes are fringes of equal thickness and are known as fizeau fringes.
- Fringes are circular due to the circular symmetry of film about the centre.
- The interference pattern observed for reflected and transmitted lights are complementary.
- At centre order of fringe is zero and it increases as one moves away.
- The rings get close as the order increases i.e. the width of rings decreases as order increases.

Summary

Newton's rings in reflected light

Path difference between the two
Reflected rays

$$\Delta_{eff} = 2\mu t \cos r + \frac{\lambda}{2}$$

Interference is maximum when

$$2\mu t = (2n - 1)\frac{\lambda}{2}$$

Interference is minimum when

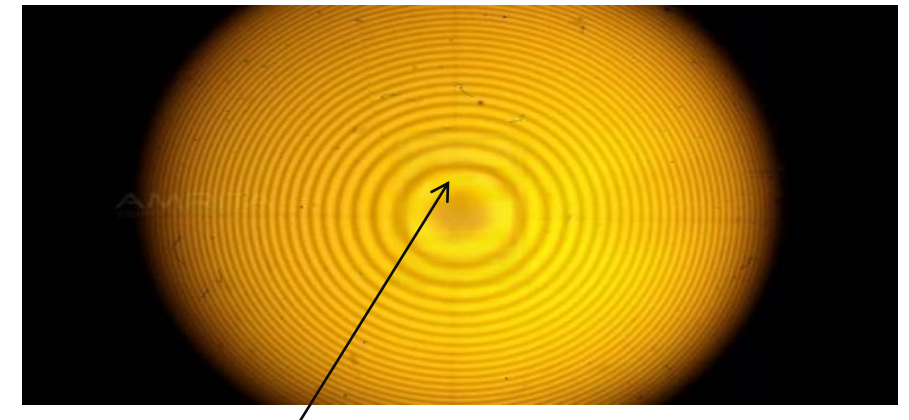
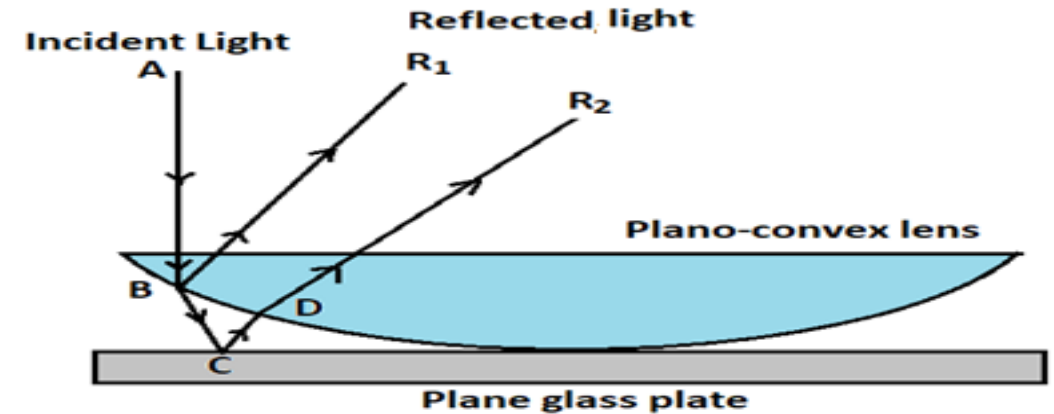
$$2\mu t = n\lambda$$

Diameter of bright ring

$$(D_n)_{bright} = \sqrt{\frac{2(2n-1)R\lambda}{\mu}}$$

Diameter of dark ring

$$(D_n)_{dark} = 2\sqrt{\frac{Rn\lambda}{\mu}}$$



Central fringe is dark

Summary

Newton's rings in transmitted light

No additional phase change of π

(or path difference of $\lambda/2$) in transmitted rays.

So path difference

$$\Delta = 2\mu t$$

Interference is maximum when

$$2\mu t = n\lambda$$

Interference is minimum when

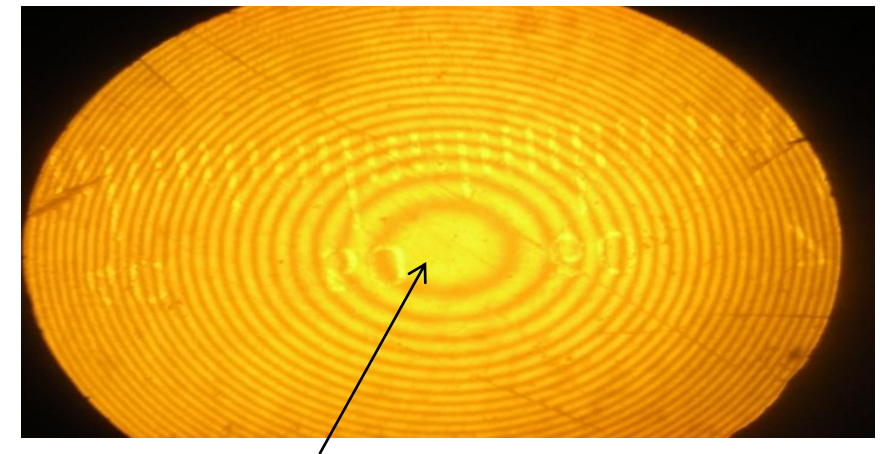
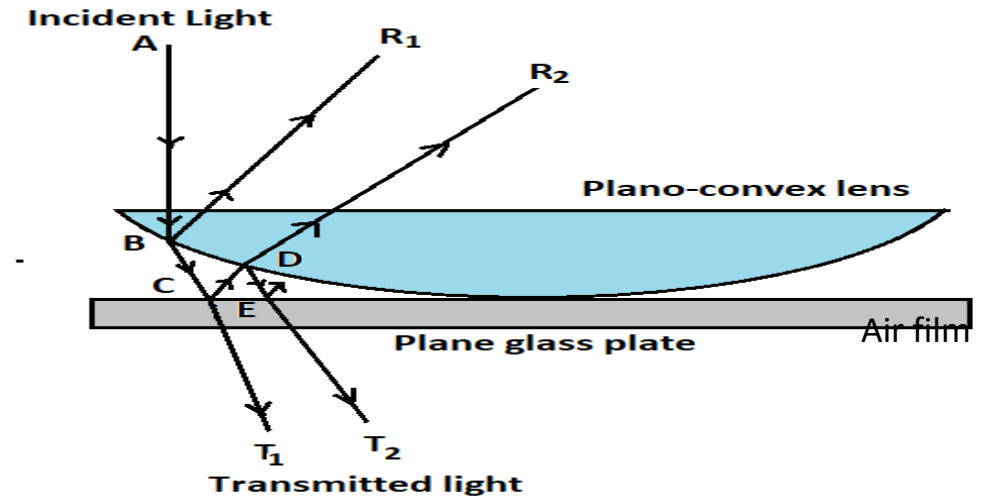
$$2\mu t = (2n-1)\frac{\lambda}{2}$$

Diameter of bright ring

$$(D_n)_{\text{bright}} = 2\sqrt{\frac{Rn\lambda}{\mu}}$$

Diameter of dark ring

$$(D_n)_{\text{dark}} = \sqrt{\frac{2(2n-1)Rn\lambda}{\mu}}$$



Central fringe is bright

Source: https://en.wikipedia.org/wiki/Newton%27s_rings

Summary

- On using plano-convex lens of small radius of curvature the percentage error in the measurement of diameter increases.
- On replacing plane glass plate by mirror fringes will disappear.
- The radii of Newton's rings decrease on raising the plano-convex lens from glass plate.
- If plane glass plate below the plano-convex lens is replaced by another plano-convex/plano-concave lens the radius of the rings will change and depend on radius of curvature of both lens.
- If plano-convex lens is replaced by plano-concave lens zero order fringe will form at the periphery of the lens and will be dark.
- Using white light instead of monochromatic light, few coloured rings are observed and after that due to overlapping of rings of higher order and different wavelengths a general illumination is obtained.
- Newton's ring experiment can be used to determine wavelength of monochromatic light and refractive index of liquid in small quantity.

Questions asked in Exams

- Q1. Why radius of curvature is large for Newton's ring experiment. Explain why Newton's fringes are fringes of equal thickness.
- Q2. Explain the formation of Newton's rings in reflected light. Why Newton's rings are circular in shape? Prove that the diameters of dark rings are proportional to the square root of natural numbers and also show that rings are not evenly spaced?
- Q3. Prove that the diameters of bright rings in reflected light are proportional to the square root of odd natural numbers and also show that rings are not evenly spaced?
- Q4. Prove that for Newton's ring experiment, the interference patterns in reflected and transmitted light are complementary to each other.

Questions Asked in Exams

Q5.Explain Newton's ring method for determining the wavelength of monochromatic light. Why is the centre of the ring dark and how can we get a bright centre? What happens to the fringe pattern when white light source is used?

Q6.How will you find the refractive index of a liquid with the help of Newton's rings?

Q7. What will be effect on Newton's rings if:

- (a) A plane mirror is placed instead of the glass plate below the plano convex lens.
- (b) A Plano convex lens of small radius of curvature is used.
- (c) The Plano convex lens is raised by height Δh from the surface of plane glass plate.

Questions Asked in Exams

- (d) A little oil ($\mu=1.65$) is introduced between the lens ($\mu=1.5$) and the glass plate ($\mu=1.75$)
- (e) A plano concave lens is used instead of plano convex lens
- (f) A plano concave lens is used instead of glass plate below the plano convex lens
- (g) If plane glass plate below plano convex lens is replaced by a rough surface

**Thank
You**

